

Speaking the Aggressor's Language: Identity and Sanctions Evasion ^{*}

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Abstract

Export controls work only if third countries decline to circumvent them. We study how language identity affects sanctions circumvention in a bystander state of the Russia-Ukraine war. In a difference-in-differences framework for Kazakhstan's trade, re-exports of sanctioned goods rose only where Russian was widely spoken among ethnic Kazakhs before 2022, with no such pattern toward other Eurasian Economic Union members. A pre-registered survey experiment identifies the mechanism at the individual level: respondents randomly addressed in Russian on the introductory page are more likely to justify the invasion and helping Russia evade sanctions, although the cue leaves their information and incentives unchanged, and the effect is specific to Russia-related attitudes. The Russian language also polarizes attitudes toward Russia after the invasion and raises online searches for pro-Kremlin news. Language identity thus shapes both the beliefs and the economic behavior of citizens in a country watching the war from outside.

Keywords: export sanctions, language, sanctions evasion, Kazakhstan, Russia-Ukraine war

JEL: F51, P45, D74, Z13

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Armed conflict reshapes economic exchange far beyond the battlefield. A growing literature shows that war cuts trade between groups whose identities it sets against one another, even in places where no fighting occurs ([Korovkin and Makarin 2023](#)). Much less is known about the mirror case, namely countries that are not party to the war at all. However, it is precisely thanks to these bystanders that the outcome of the economic pressure of modern warfare is increasingly being determined. Export controls bind only if firms in neutral third countries decline to reroute the goods those controls cover, and many do not decline: rerouting through friendly neighbors offsets a large part of the intended decline in the sanctioned country's imports ([Chupilkin et al. 2026](#); [Egorov et al. 2026](#)). Who supplies a sanctioned aggressor, and why, is the question of this paper.

The standard answer is profit. Sanctions make goods scarce in the target country, and circumvention is arbitrage: whoever can collect the rents with minor risk would do that. We ask whether identity is also part of the answer, i.e., could citizens of a bystander country help a former colonizing power wage war because they have come to identify with it? In an attacked country, war forces a choice; citizens rally around the flag, and the population consolidates around the nation even where prior ties to the aggressor were deepest ([Abramenko et al. 2026](#)). A bystander faces no such imperative. What it inherits instead is the pull toward the former metropole, carried above all by language, which in much of the post-Soviet world still groups citizens into information worlds and identities.

Establishing that this pull is causal is hard, and this is the second gap we address. Language is not randomly assigned. Regions where a former colonizer's language is spoken also lie closer to it, are more likely to share the ethnic identity, and contract with its firms face lower transaction costs, so a correlation between language and trade with the metropole is consistent with identity, geography, and pure transaction costs alike. Isolating the language identity effect requires variation in identification itself, holding information, incentives, and every fixed trait constant. This paper provides such variation, and shows that language moves attitudes toward a foreign power's war and toward evading the sanctions against it.

Kazakhstan after February 2022 is a natural laboratory for this investigation. First, the country is not at war but shares a border, a customs union, and a colonial past with Russia, so we observe how citizens respond when war makes an identity salient without the destruction

and mobilization that confound studies of countries at war. Second, more than thirty years after independence, and although the state language is Kazakh, Russian remains the everyday language of a large share of the population, including many ethnic Kazakhs. Speaking Russian and being ethnic Russian therefore come apart, which enables the identification of the standalone effect of language identity. Third, every region of the country faced the same post-invasion Russian demand, the same customs union, and the same profit opportunity, so uptake can be compared across regions holding the incentive fixed. Finally, Kazakhstan became a re-exporter for sanctioned goods reaching Russia in defiance of Western export controls, and its unique transaction trade records make that sanction circumvention observable at the sub-national level.

We bring four data sources. The first is the universe of Kazakhstan’s trade transactions with the Eurasian Economic Union (EAEU) from January 2021 through September 2023, aggregated to the region, month, product-class, and destination level and merged with Russian-language penetration among ethnic Kazakhs from the 2021 Census. The second is an original, pre-registered online survey experiment of 5,025 respondents fielded in November 2023, which randomizes the language of the introductory page before any question is asked. The third is eight waves of the Central Asia Barometer (CAB) spanning 2020 to 2023, four fielded before the full-scale invasion and four after it. The fourth is regional online search data on news sources.

Our main result concerns trade. Before the invasion, exports of the goods that Western countries would later sanction – electronics and machines – were small and flat, at 2.4 million USD per region-month against 8.4 million for all other goods. After February 2022, they rose by 10.5 million USD per region-month, more than quadrupling, while exports of the same product classes to the other EAEU members rose by only 0.6 million. Domestic production of these goods did not move, which identifies the flows as re-exports. We then estimate a triple difference-in-differences in which treatment exposure is the regional share of Russian-speaking ethnic Kazakhs from the pre-invasion Census, and we absorb post-invasion changes in border regions through a border-by-post interaction. The invasion by itself does not move these exports: in a counterfactual region with no Russian-speaking Kazakhs, the estimated effect is zero. The entire increase loads on the language gradient. Moving from the least to

the most Russian-speaking region, from 27 to 66 percent, raises export growth of sanctioned goods to Russia by a further 8.7 million USD per region and month. No such gradient appears for the same goods shipped to other EAEU destinations.

Even though the parallel trends assumption holds, the trade data alone cannot separate the Russian language penetration effect from other factors that correlate with regional language patterns pre-war and affect the sanctions circumvention since 2022.¹ Our survey experiment provides that separation. Respondents who clicked our online ad were shown a consent page in a randomly assigned language, Kazakh or Russian, and were free to switch the language at any time. The cue works as a first stage on identity itself: being addressed in Russian raises the probability of completing the entire survey in Russian by 22 percentage points, a revealed choice rather than a stated one, and it also raises reported use of Russian with friends and for news, though not at home, where language is most habitual. The cue then moves attitudes. Respondents assigned the Russian introduction are 3 percentage points more likely to consider helping Russia evade Western sanctions justified, 4 percentage points more likely to consider the invasion justified, and hold a more favorable opinion of Russia. Instrumenting the endogenous survey language with the randomized assigned language, we show that Russian-language identity raises justification of sanctions evasion by 15 percentage points and of the invasion by 18 p.p. Crucially, the effect is specific to Russia-related affairs: the same cue does not affect opinions of China and the United States, trust in the national government, the justifiability of protest participation, and the justifiability of overworking.

Two further pieces of evidence reveal the *polarization* pattern induced by the Russian-language penetration after February 2022 and explain why it is sustained. For the eight CAB repeated cross-section waves covering 2020-2023, using difference-in-differences, we show that respondents from regions with more Russian-speaking Kazakhs were, if anything, slightly *less* favorable toward Russia before the invasion. The positive effect of the Russian-language penetration appears in the first wave after February 2022 and persists through late 2023 even though we control for individual characteristics including ethnicity. Nothing comparable happens for opinions of the United States or China. Yandex online search data then show that

¹However, our additional specification shows that the main result holds when we extend the main specification with the ethnicity-by-post interactions.

Russian-language penetration increased pro-Kremlin news search: moving from the least to the most Russian-speaking region raises searches for war-related keywords by 13.3 per 1,000 inhabitants, with comparable increases for searches of pro-Kremlin aggregators and outlets, and only a small increase in searches for Western outlets.

We consider and rule out the main alternative explanations. In the trade results, the proximity and ethnicity are absorbed by the border-by-post and ethnicity-by-post interaction. A general wartime reorientation of commerce is inconsistent with the flat response of exports to other EAEU members. Russian as less costly contracting language predicts the same trade pattern after February 2022 but not the rest of the evidence: Russia-related attitudes should not respond to a randomized introductory language, yet they do; and language should not polarize attitudes toward Russia, while it does.

Related literature and contribution. Our paper makes four contributions. First, we contribute to the economics of identity. Since [Akerlof and Kranton \(2000\)](#) placed group identity in the utility function, this literature has treated identity as a choice: people weigh a group’s status against their perceived distance from it ([Shayo 2009, 2020](#); [Gidron and Shayo 2026](#)), and they respond to incentives by reclassifying their race ([Dahis et al. 2019](#)), choosing their children’s ethnicity ([Jia and Persson 2021](#)), or expressing identity through what they consume ([Atkin et al. 2021](#)). We contribute a randomized cue in the field that shifts identification in a setting where the identity at stake carries material consequences, and a design – randomizing the language in which respondents are addressed, before any question is asked – that is cheap and portable to any multilingual setting, adding to the measurement toolkit surveyed by [Rohner and Zhuravskaya \(2025\)](#). The experiment also speaks to *which* trait carries an identity, a question on which ethnicity and culture are known to come apart ([Desmet et al. 2017](#)). Literature on Russia’s influence over the post-Soviet space attributes loyalty to Russian ethnicity ([Peisakhin and Rozenas 2018](#); [Korovkin and Makarin 2023](#); [Larreguy and Martinez 2024](#)), while research on the causal effect of former colonial languages is scarce. [Pérez and Tavits \(2017, 2019\)](#) use the randomized survey language design in Estonia, but their sample is selected only for bilingual respondents, and they force the respondents to complete the survey in a language they are assigned. Moreover, their outcomes are beliefs related to the grammatical traits of the assigned language without connection to the political origin of the

language.² In all our empirical evidence, we are able to separate ethnicity from the language. Moreover, the survey experiment randomizes language without forcing it and identifies the language effect on attitudes toward foreign power.

Second, we contribute to the literature on nation-building and on how wars redraw group boundaries (Rohner and Zhuravskaya 2023, 2025). Attacked Ukraine consolidated: the invasion raised the salience of a trait every resident shares, the side of the interstate boundary, and pulled all groups toward the nation (Abramenko et al. 2026). Kazakhstan polarized: the war raised the salience of a trait its residents sit at opposite ends of, and pulled them apart. Consolidation and polarization are thus not competing theories but one mechanism under different patterns of exposure, and the direction depends on whether the trait a war makes salient unites or divides the population. This also connects to work on identities formed by dividing events such as Brexit (Hobolt and Tilley 2026) or vaccination status (Bor et al. 2023; Henkel et al. 2023; Wagner and Eberl 2024), which display the full repertoire of affective polarization (Mason 2015; Iyengar et al. 2019). In our case, a war abroad polarized a bystander society along a cleavage that predates it by a century, with no vote or referendum compelling anyone to take a side.

Third, we connect identity to international trade and to the effectiveness of sanctions (Drezner 2024). A shared language with a former colonial power is ordinarily an asset, lowering the cost of communication and contracting and raising trade (Melitz 2008; Melitz and Toubal 2014; Guiso and Makarin 2020; Gurevich et al. 2025). We show that under sanctions, the same tie carries loyalty, making our results the behavioral mirror image of Korovkin and Makarin (2023): identity not only severs economic ties under conflict, but it can also intensify them among groups sharing the same identity for conflict bystanders. This provides a micro-level foundation for the role of Central Asia in Russia’s evasion of Western sanctions (Chupilkin et al. 2026; Babina et al. 2023; Egorov et al. 2025, 2026). That literature models circumvention as arbitrage and asks how much leaks; we ask who does the leaking. The opportunity was identical across Kazakhstan, yet the rents were collected only where language ties people to Russia, and left uncollected where it does not. Forgoing profit, rather than taking it, is what

²On the non-causal effect and persistence of imperial languages, see Phillipson (2008) and Crystal (2012) on the British empire, Neathery-Castro and Rousseau (2005) and Vigouroux (2013) on the French, and Mar-Molinero (2000) and Stavans (2021) on the Spanish.

makes identity economically essential (Atkin et al. 2021). It also carries a policy implication: export controls consider routes, products, and firms, but the supply of willing intermediaries also follows identity, a margin such controls are not built to consider.

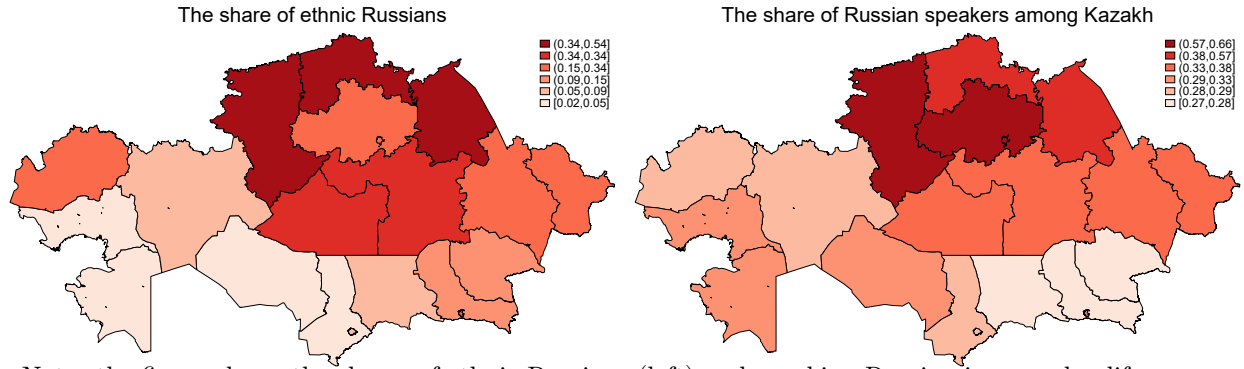
Fourth, we speak to the literature on media as an instrument of cross-border influence. Where the state controls the media environment, framing shifts attitudes regardless of predispositions (Enikolopov et al. 2011; Zhuravskaya et al. 2020; Pan et al. 2022). Where audiences can choose, effects run along prior loyalties: Russian TV raised pro-Russian voting in Ukraine among viewers already predisposed toward Russia while backfiring among the rest (Peisakhin and Rozenas 2018), independent television inside Russia polarized rather than persuaded (Enikolopov et al. 2026), and Beijing-backed media in Taiwan did the same (Kao 2026). This literature studies what media do to a given audience. We study how that audience forms upstream of media consumption. Language steers citizens toward the foreign power’s outlets, and because our cue moves attitudes with no media exposure at all, media in our setting enters downstream of identity, amplifying and stabilizing a shift that identity choice triggered.

The remainder of the paper proceeds as follows. Section 1 provides background and context, and Section 2 develops the conceptual framework and derives its predictions. Section 3 describes the data. Section 4 presents our main trade results. Section 5 turns to the mechanism: Section 5.1 reports the survey experiment, Section 5.2 traces the dynamics of attitudes in the CAB panel, Section 5.3 examines online search behavior, and Section 5.4 rules out alternative explanations. Section 6 concludes.

1 Background and Context

One of the successor states of the Golden Horde, the Kazakh Khanate was gradually conquered and absorbed by the Russian Empire from the late 18th to the mid-19th century. Following the Russian Revolution, the territory became part of the Soviet Union. After several territorial reorganizations, Kazakhstan’s modern borders were established in 1936 with the formation of the Kazakh Soviet Socialist Republic. Kazakhstan declared its independence from the Soviet Union on December 16, 1991 and has since been an independent state. Kazakhstan, like other former colonies, has seen real decolonization efforts, though its stance toward Moscow has long remained ambivalent.

Figure 1: Regional heterogeneity by ethnicity and language use



Note: the figure shows the shares of ethnic Russians (left) and speaking Russian in everyday life among ethnic Kazakh (right) by regions of Kazakhstan according to the Census 2021. Regions are three major cities – Astana, Almaty, and Shymkent – and 17 other sub-national regions. The Pearson correlation between the regional share of ethnic Russians and the share of Russian speakers among Kazakhs is 0.64.

As a result of two centuries of Russian colonization, a significant Russian minority continues to live in Kazakhstan. Although Kazakh is the official state language, Russian also holds the status of an official language, and a substantial portion of the population continues to speak Russian daily. According to the 2021 Census, the share of ethnic Kazakhs is 70%, and that of ethnic Russians is 15.5%. Moreover, among ethnic Kazakh, the share of people speaking and reading Russian is 79.5%, and using Russian in everyday life is 36.5%.³ Figure 1 illustrates the distribution of ethnic Russians and Russian speakers among Kazakhs by region, showing that Russian speakers are concentrated in the north of the country and in the two main cities – Astana and Almaty. We show throughout the paper that this regional variation in language use determines political opinions and economic behavior, including trading sanctioned products with Russia.

Since the start of Russia’s war in Ukraine in 2014, the Kazakh government has been careful to maintain a low profile with respect to the conflict, avoiding too extensive coverage of the conflict in the official media and employing “strategic silence” (Dadabaev and Sonoda 2023), in order not to antagonize neither Russia nor the West, even though at times concerns and criticism about the war are voiced.⁴ As a result, almost all of the Kazakh- and Russian-language media published within Kazakhstan take either a neutral or a moderately critical

³Results of the National Population Census 2021

⁴Steppe change: How Russia’s war on Ukraine is reshaping Kazakhstan. EC on Foreign Relations, 2023.

stance towards Russia when covering the conflict.⁵ However, at the same time, Russian-language media from Russia is widely available in Kazakhstan. Especially when searching for information on the internet, the probability that Russian speakers will end up on a site from Russia featuring pro-Kremlin content is high, while Kazakh speakers are more likely to find content that is more neutral with respect to Russia’s invasion of Ukraine.

While the Kazakh government has been careful to maintain a neutral stance towards the conflict in its official statements, economically, Kazakhstan has taken a pro-Russian stance. Exports from Kazakhstan to Russia have increased significantly since February 2022. However, as illustrated by Figure 2, this is *only* the case for electronic goods and machines, i.e. products on which Western countries imposed sanctions after Russia’s full-scale invasion of Ukraine in February 2022, while exports of other goods have remained remarkably stable.⁶ After February 2022, Kazakhstan reduced import of sanctioned products from Russia (Figure A2), slightly increased their import from other EAEU countries (Figure A3), and substantially increased the import of sanctioned products from the rest of the world (Figure A4), without significant changes in domestic production (Figure A5). Such cases of sanctions circumvention were also revealed by investigative journalists,⁷ and US Secretary of State Antony Blinken visited Kazakhstan in February 2023 to discuss specifically the issue of sanctions circumvention.

2 Conceptual Framework

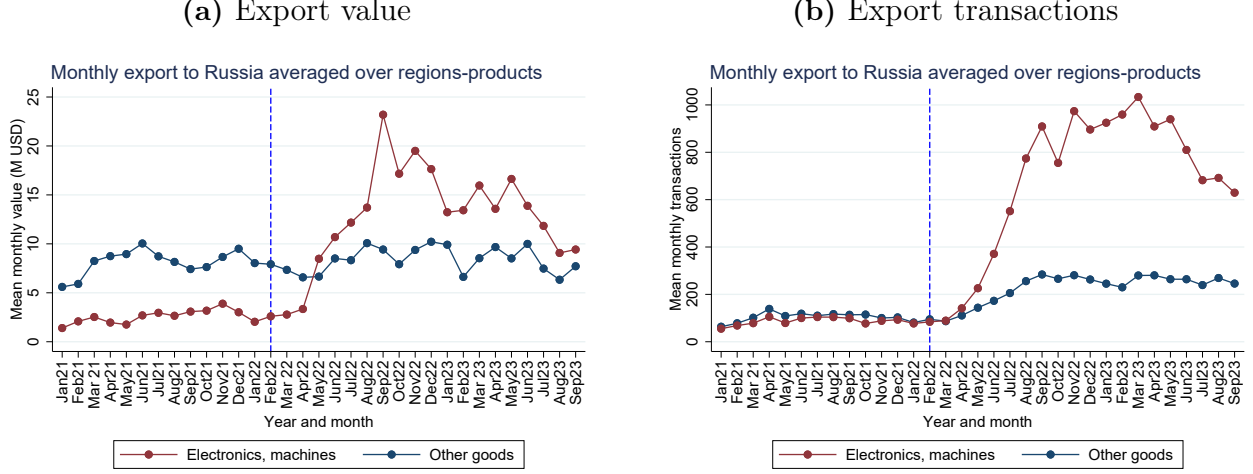
We start with a conceptual framework that predicts behavior based on the identity choice. We adopt the social identity model of Shayo (2009, 2020) in the form that Abramenko et al. (2026) develop for wartime Ukraine. We also inherit their notation and definitions while adjusting the interpretation for a bystander country – like Kazakhstan. A resident i chooses

⁵This is also a result of Kazakh media being largely state-controlled. According to Freedom House, “the dominant media outlets in Kazakhstan are either in state hands or owned by government-friendly businessmen”.

⁶Figure A1 of the Online Appendix A shows Kazakhstan’s export value and transactions to other EAEU countries, excluding Russia. While there is a surge after February 2022, it is of a smaller order.

⁷See. e.g. Kazakhstan Has Become a Pathway for the Supply of Russia’s War Machine. Here’s How It Works. Organized Crime and Corruption Reporting Project, 2023. or Now through Kazakhstan: devices from the United States, Germany and France, which can be used for electronic warfare, continue to go to Russia. The Insider, 2024. or Shooting Ukraine in the back: Sniper rifles and ammunition from the EU and U.S. are being supplied to Russia despite sanctions. The Insider, 2024. or Russian businessman extradited to Estonia for selling \$50 million worth of foreign equipment to Russian defense enterprises. The Insider, 2025.

Figure 2: Export from Kazakhstan to Russia



Note: the figure shows the monthly value of export in millions USD (left) and the number of export transactions (right) of Kazakhstan to Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3 for the international trade data.

which of two identities to identify with, $j \in \{N, R\}$, the Kazakh nation N or the transnational Russian-speaking world R . Identification is endogenous and may be linked to the individual's legacy and current circumstances. The value of identifying with group j is

$$I_{ij} = (1 - \beta) S_j - \beta d_{ij}, \quad (1)$$

where S_j is the group's *status*, the esteem it enjoys in the wider social order, d_{ij} is i 's perceived *distance* from the group's typical member, and $\beta \in (0, 1)$ sets the trade-off between the two. People value belonging to a group more when the group is well regarded and when its members are similar to themselves. Distance aggregates over three traits, $k \in \{\text{language, ethnicity, culture}\}$,

$$d_{ij} = \sum_k w_k \delta_k(i, j), \quad (2)$$

where $\delta_k(i, j)$ is i 's gap from group j 's prototype on trait k and $w_k \geq 0$ is an *attention weight* measuring how much trait k is currently on people's minds. These weights are the active part. An event can thrust a trait into focus and raise its weight without moving anyone's actual position on it, as when an ethnic war raises the weight on ethnicity (Sambanis and Shayo 2013). Identity is thus governed by three forces, namely group status, trait salience (weights),

and the person’s own position in trait space. Appendix A gives a numerical example.

Abramenko et al. (2026) applies this framework to Ukraine, where the 2022 invasion moved status and salience in the same direction. Successful resistance and international solidarity raised S_N while military setbacks and isolation lowered S_R . As the attack struck the country as a whole and explicitly denied that a Ukrainian nation exists, the trait thrust into focus was the side of the interstate boundary, a trait every resident shares. Raising the weight on a shared trait lowers d_{iN} and raises d_{iR} for everyone at once, most sharply for Russian-speakers. The framework predicts *consolidation* and the authors find districts of more ethnic Russians converging toward the national mean and residents switching to Ukrainian in language, music, and baby names. Our contribution is to apply the framework to a *bystander* country whose citizens have the identity R available to them. A bystander lacks precisely the forces that consolidated Ukraine. No resistance lifts S_N , whatever the war does to S_R is common to every resident and cannot pull two groups in opposite directions, and ethnic distances are large and fixed, as ethnic Kazakhs identify as Asian rather than Slavic. What the war moves is the salience of language, because Russia declared the invasion a war for the “Russian world” and the protection of Russian-speakers wherever they live.⁸ Language, unlike the interstate boundary in Ukraine, is a trait on which residents sit at opposite ends, so the higher weight w_{lang} magnifies gaps that may already exist, making R better to identify with for a Russian-speaker and worse for a Kazakh-speaker. Where the attacked country consolidates around a trait its residents share, the framework predicts that a bystander *polarizes* along the trait that divides its residents, with conformity following downstream as those newly identifying with R consume Russian-language media, defend Russia’s war, and sustain the trade that supplies it. This logic yields five predictions that we take to the data.

Prediction 1 (Status). After February 2022, the average opinion of Russia should fall, because the war lowers Russia’s status S_R and this component of equation (1) enters identically for every resident. Being common to all, it shifts the level of pro-Russian sentiment without distinguishing between language groups. The fall should be specific to Russia, since the shock is to S_R and not to how residents regard third countries such as China or the United States.

⁸Abramenko et al. (2026) note that in principle their mechanism “could instead have sharpened internal ethnolinguistic divisions under different patterns of exposure.” The bystander, exposed to the war’s framing but not to its violence, is that case.

Prediction 2 (Polarization). After February 2022, an attitude gap between Russian- and Kazakh-speakers should widen, on top of the common decline in Prediction 1. Before the war, language carried little attention weight, so it should predict attitudes only modestly. After the invasion, because the operative trait is language and not ethnicity, the gap should appear even among Russian-speaking ethnic Kazakhs, a group that any ethnicity-based account would place firmly with N .

Prediction 3 (Identity activation). The mechanism runs through perceived distance, so an exogenous cue that lowers the felt distance d_{iR} should shift attitudes toward Russia’s positions even without any new information about the war. The shift should be specific, appearing only on topics that map onto the divide between N and R .

Prediction 4 (Behavioral conformity). Identity choice is consequential, so it should surface in survey responses and behavior. The most Russian-speaking regions should sustain trade with Russia after the invasion, including in sanctioned goods, since supplying the group one identifies with is conformity to its interests. The sanctions themselves create scarcity rents in Russia that raise the material return to this trade for every region alike. What the model adds is that uptake of this common profit opportunity varies with identification, concentrated where the Russian-language tie is strong and absent where it is not. Conversely, for residents who identify with the Kazakh-speaking majority, conformity means refusing this trade, so they should leave the rents uncollected even though the same profits are available to them. Identification thus predicts both who collects the rents and who declines them, a feature that Section 4 tests directly.

Prediction 5 (Reinforcement and persistence). Demand for Russian-language media should rise as an expression of conformity during high salience moments. Consuming that media in turn lowers d_{iR} , by shifting positions on cultural traits toward the Russian prototype, and raises perceived S_R through propaganda.

3 Data

To understand how linguistic identity affects attitudes and trade patterns toward Russia, we combine several data sources. First, we use transaction-level trade data between Kazakhstan and the Eurasian Economic Union (EAEU) for the period 2021 to 2023 and merge it with

the regional share of Russian-speaking Kazakhs from the 2021 Census. It enables us to investigate how Russian-language penetration affects the re-export of sanctioned products to Russia after February 2022. Second, we conducted an original online survey in November 2023 that embeds a pre-registered language-randomization experiment, allowing us to test whether Russian-language identity causally shifts attitudes toward the war and sanctions circumvention. Third, using eight waves of a nationally representative telephone survey conducted by the Central Asia Barometer in Kazakhstan in 2020–2023, we test whether pre-war Russian-language penetration polarizes attitudes toward Russia in Kazakhstan. Finally, using data on online searches on Yandex, we test whether this divergence in political views can be partially explained by differences in online news search after February 2022.

3.1 International trade

We use data from international trade between Kazakhstan and Russia to study the effect of Russian-language penetration on re-export activity. Our dataset was collected from the Bureau of National Statistics of Kazakhstan (www.stat.gov.kz). It consists of all transactions between Kazakhstan and the countries of the Eurasian Economic Union for the period January 2021 – September 2023. Each transaction contains information on the traded product classification code (TNVED, coinciding with the HS codes), the trading firm’s region in Kazakhstan, the trading partner’s country, the declared product value (in USD and KZT), and the gross weight. Focusing on export activity from Kazakhstan, encountering 811,000 individual transactions, we classify each of them into five product classes: (i) electronics and machines, (ii) food, (iii) textiles, (iv) metals, and (v) others.⁹ Most of the products from class (i) are under Western sanctions, prohibiting their re-export from Kazakhstan to Russia. We therefore call them *sanctioned products*, even when discussing the period before 2022. Among all export transactions, 78.8% are to Russia and 17.4% to Kyrgyzstan. We aggregate the export data, calculating the export value (in millions of USD) and the number of transactions at the regional level (17 regions)¹⁰, monthly (33 months), product class (5 classes) levels, and desti-

⁹We allocate the 2-digit TNVED product classifications as follows: electronics (85,88,90,91), machines (84,86,87,89), food (01-24), textiles (41-43, 50-67), metals (72-83), with “others” covering the rest.

¹⁰In 2021, there were 17 regions; in 2022, three regions were split. As we merge the trade data with the Census 2021 data, all export transactions from new regions in 2022 are assigned to their 2021 parent regions.

nation country (Russia vs. other EAEU countries). Figure 2 shows the dynamics of exports from Kazakhstan to Russia averaged over regions and product classes for electronics/machines and other products (product classes (ii)-(v)). Figure A1 of Online Appendix A shows these dynamics for exports to other EAEU countries. Finally, the Bureau of National Statistics of Kazakhstan also provides industrial production output data for each region and month for 36 OKED product categories, which we aggregated into the same 5 product classes.¹¹ Figure A5 of Online Appendix A shows the dynamics of Kazakhstan’s production averaged over regions and product classes for electronics/machines and other products. Notably, there is no significant change in sanctioned product production, though.

3.2 Language experiment in an online survey

We have conducted a pre-registered¹² online survey experiment fielded in November 2023 via NAC Analytica, a leading public opinion pollster in Kazakhstan.¹³ Participants were recruited through advertisements on Instagram and compensated 700 Tenge (approx. 1.50 USD).¹⁴

People who clicked the survey ad on Instagram were directed to a survey platform, where the introductory page with the consent form was displayed in a *randomly assigned* language, Kazakh or Russian. This randomization is the centerpiece of our design. Respondents were free to switch language at any point of the survey, including the introductory page. In practice, those who switched did so immediately on the introductory page or the first demographic question, and completed the survey in their chosen language.¹⁵ Table C2 of Appendix C confirms that the randomization was successful: across all demographic characteristics, Cohen’s d is below 0.1.

We have several, potentially endogenous, measures of linguistic identity. The first is the

¹¹We allocate the 2-digit OKED product classifications as follows: electronics (26,27), machines (28,29,30), food (10,11,12), textiles (13,14,15), metals (24,25), with “others” covering the rest.

¹²https://aspredicted.org/YPM_LSH

¹³<https://nacanalytica.com/en/>

¹⁴Quality verification included an attention check on respondents’ ages, a minimum response time, and a restriction on duplicate participation by IP address and phone number. Out of 28,201 users who clicked the survey ad on social media to read the consent form, 5,025 completed the survey and passed all quality checks as unique respondents; most who dropped out did so on the consent page or during the initial demographic questions.

¹⁵We tracked the language for each question.

language in which a respondent *completes* the survey, which we consider a *revealed* preference for a language (Atkin et al. 2021). The second set of measures includes *stated* language use in different conditions. We ask which language respondents use most often (i) at home, (ii) with friends or colleagues, and (iii) for news about political and economic events abroad.¹⁶

Our outcomes fall into two groups. The first includes *Russian affairs*: (i) whether respondents consider it justified to help Russia evade Western sanctions,¹⁷ (ii) whether they consider Russia’s invasion of Ukraine – its “Special Military Operation” – justified,¹⁸ and (iii) the overall opinion of Russia. The second group of outcome variables includes *non-Russian affairs*: (iv) opinions of China and the United States, (v) trust in government, (vi) the justifiability of participating in protests for political change, and (vii) whether working more than 50 hours a week is justified. All questions offered a *Prefer not to answer* option.¹⁹ Table C1 reports summary statistics for the survey sample. The share of respondents who favor sanctions circumvention is 30%, and the share who justify the invasion of Ukraine is 26%. The latter is similar to the share justifying the invasion from the CATI survey of the Central Asia Barometer, which we present next.

3.3 Central Asia Barometer

Since June 2017, the Central Asia Barometer (CAB) has been conducting public opinion surveys in the four Central Asian republics of Kazakhstan, Kyrgyzstan, Uzbekistan, and Tajikistan twice a year in April-June and September-November. For our study, we use waves of the 2020–2023 CAB for Kazakhstan.²⁰ All eight survey waves include questions about attitudes toward Russia, the United States, China²¹ and a block of demographic questions

¹⁶We also asked about the set of languages respondents know well.

¹⁷“In your opinion, is helping Russia avoid Western sanctions generally justified or not justified?”

¹⁸“In your opinion, is (Russia’s Special Military Operation in Ukraine/ invasion of Ukraine) generally justified or not justified?”. The wording “Special Military Operation in Ukraine” versus “invasion of Ukraine” was itself randomized to balance framing effects.

¹⁹The survey also fielded a set of list experiments for (i), (ii), (vi), (vii). The placebo list (vii) showed a significant design effect; so we rely on the direct questions and do not analyze the lists in this paper.

²⁰Since 2020, these have been CATI surveys with randomly generated mobile phone numbers drawn from the mobile operators’ number ranges in the respective countries. The last available wave is in Fall 2023. For more details, see <https://www.ca-barometer.org/en>.

²¹The question is: “Thinking about other countries, please tell me if you have a very favorable (4), somewhat favorable (3), somewhat unfavorable (2), or very unfavorable (1) opinion of (Russia/the United States/China).”

including ethnicity, language spoken at home, and the language of the interview.

Table 1: Descriptive statistics for the language, ethnicity, and opinion of countries

Year	Obs.	Home lang.	Russian (%)	Ethnicity	Favorable opinion of (%)		
			Survey lang.		Russia	US	China
2020	3,500	49.9	85	25	77.1	46.5	46.5
2021	3,500	47.3	78.5	22.2	70.5	51.9	44.9
2022	3,020	41.1	70.3	17.9	50.9	49.4	51.5
2023	3,000	45.3	70.6	20.5	57.1	51.4	60.3

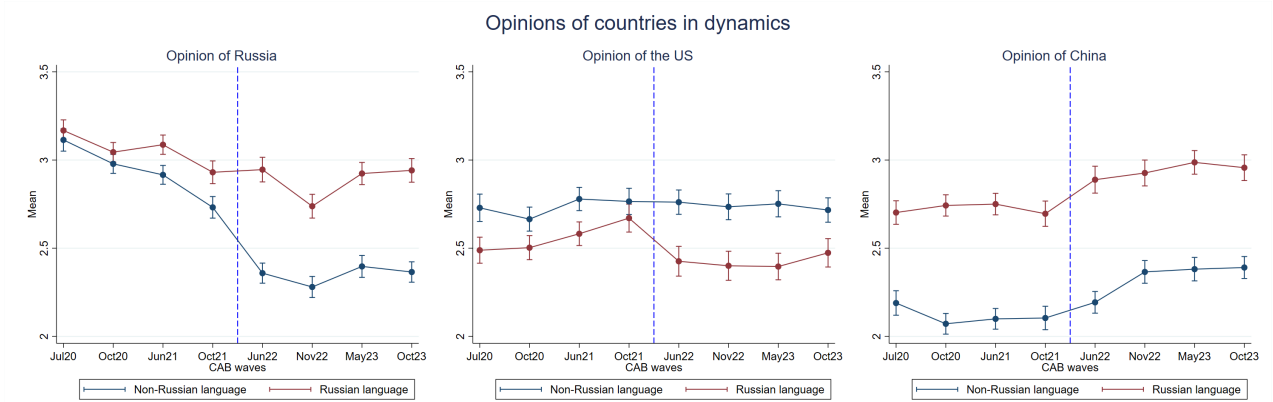
Note. The table uses CAB survey waves for 2020-2023. The “Favorable opinion” columns show the percentage of “Very favorable” or “Somewhat favorable” opinions. Table A1 shows descriptive statistics for the war attitudes and responsibility attribution.

Table 1 presents descriptive statistics for language, ethnicity, and attitudes by year. The share of respondents who speak Russian at home ranged from 41% to 50% and is more than twice the share of ethnic Russians. Notably, the share of respondents completing the survey in Russian dropped from 85% in 2020 to 70.6% in 2023. Holding a favorable opinion of Russia fell from 77% in 2020 to 57% in 2023, with a minimum in 2022. Favorable views of the US have remained remarkably stable, while views of China have improved since 2022.

Figure 3 shows the dynamics of opinions toward Russia, the US, and China, split by home language. There was a broadly similar downward trend in opinions toward Russia, with a minor language gap before 2022; the gap widened substantially after 2022. Similarly, US opinion was relatively stable with a minor language gap before 2022; the gap widened substantially thereafter. China’s opinion has evolved in parallel for the two languages before and after 2022. Figure A6 shows similar dynamics split by the survey language.

The waves starting from 2022 also include several questions related to Russia’s full-scale invasion of Ukraine, of which we use the following two in the Appendix: (i) “*In your view, who is mainly responsible for the situation in Ukraine?*”, and (ii) “*To what extent do you think Russia’s special military operation in Ukraine is justified or unjustified?*”.

Figure 3: Dynamics of Opinions of Russia, the US and China by home language.



Note. The figure shows the mean trends in opinions across different countries by **home** language (Russian or Kazakh), based on CAB data for 2020-2023. Figure A6 shows the dynamics, split by the **survey** language.

3.4 Yandex Search Data

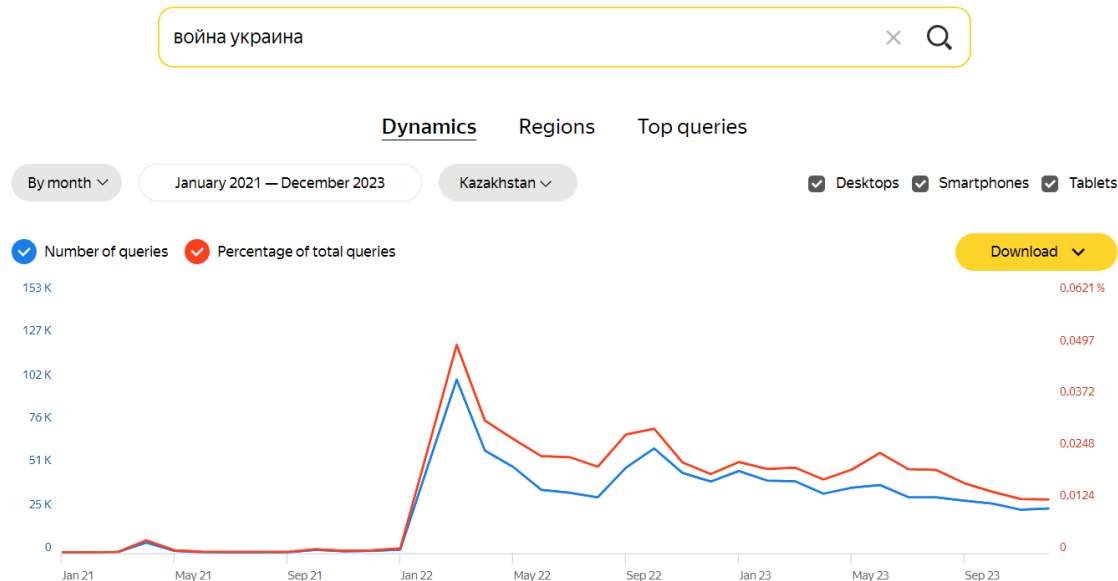
We use Yandex search data to study how regional Russian-language penetration affects online search behavior for information about the war.²² The data is taken from wordstat.yandex.kz. It shows the overall number and percentage of searches for a specific phrase per month and region. Figure 4 shows a substantial spike in searches for the phrase “war in Ukraine” in Russian in February 2022, with a smaller spike around the Russian mobilization in September 2022 and a stabilization after that.²³ Figures A9, A10, A11 and A12 in Appendix A document that interest in internet news more broadly also spiked in February 2022. In addition, we document that while searches for information from pro-Kremlin news aggregators Yandex Dzen and Yandex News (Figures A10 and A11 of Appendix A) were significant and growing after February 2022, there is only a minor increase in searches for Euronews, the most popular

²²Yandex has a [market share](#) of roughly 15% in Kazakhstan, whereas Google handles the vast majority of searches. However, Google search data is not accessible in a systematic and reliable manner due to several limitations. Google Trends relies on a subsampling methodology, meaning the provided statistic is based only on a subset of searches, making the search statistics inconsistent over time. Additionally, Google Trends provides only data on the relative popularity of search terms, making it difficult to assess actual search volumes or to make accurate comparisons over time or across keywords and regions. In contrast, Yandex Wordstat offers absolute search counts, providing more stable and reliable data for analysis.

²³Figure A7 of Appendix A presents the search results for the same phrase, “war in Ukraine”, in Kazakh. The number of searches does not exceed 130 per month, while queries in Russian reach tens of thousands, justifying our exclusive focus on search keywords in Russian. Moreover, Google Trends was unable to estimate the phrase’s popularity in Kazakh, displaying a message about insufficient data, as shown in Figure A8 of Appendix A.

pro-Western news channel in Kazakhstan (Figure A12 of Appendix A).

Figure 4: Search for “War Ukraine” on Yandex



4 The Effect of Russian Language on Sanctions Circumvention

In this section, we study how Russian-language penetration affects the export of sanctioned goods from Kazakhstan to Russia and to other EAEU countries. Our aggregate export data includes 5610 observations at the region, month, product-class, and destination levels (17 regions, 33 months, 5 product-classes, 2 destinations). This section tests Prediction 4 of our conceptual framework. If identification with the Russian-speaking world increased following the full-scale invasion in 2022, it should surface in conformity behavior, and the most Russian-speaking regions should increase trade with sanctioned products that Russia demands.

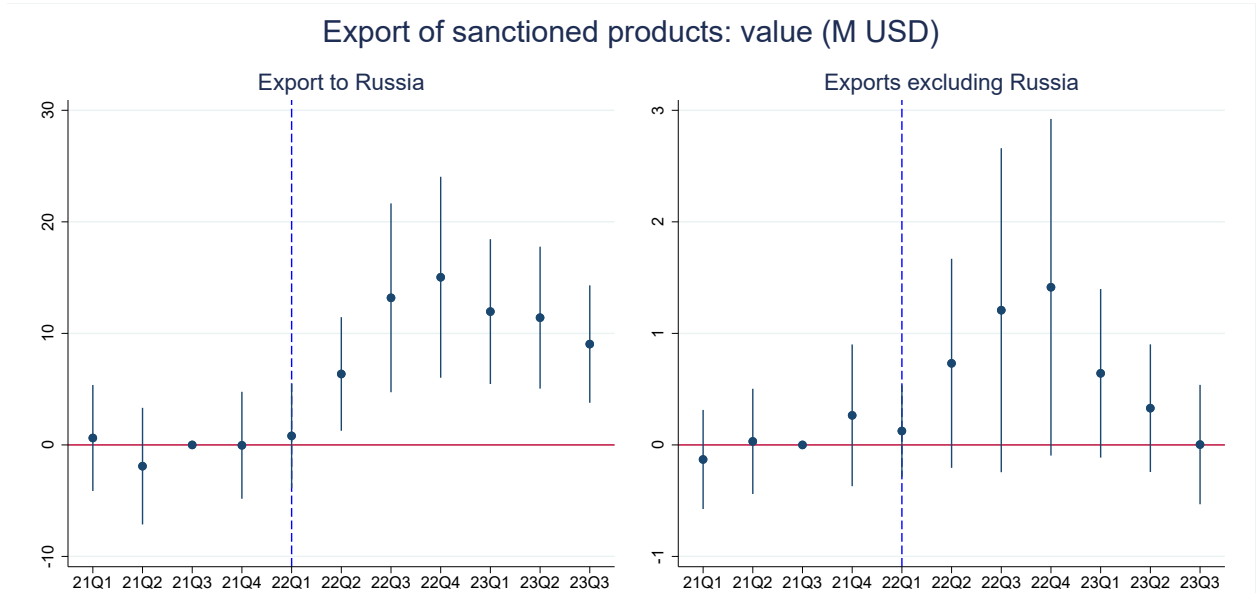
Figure 2 shows that before the full-scale invasion, exports of subsequently sanctioned goods from Kazakhstan to Russia were small and relatively stable. While non-sanctioned goods were exported to Russia at the level of 8.4 million USD on average per month-region, the export of goods that were subsequently sanctioned stood at around 2.4M USD per month-region. In the months after the start of the full-scale invasion, these exports increased multiple times, reaching a spike of 20M USD per month-region in September 2022, while the export of non-sanctioned goods remained stable at pre-invasion levels.

To begin with, we estimate the average effect of the full-scale invasion and the sanctions on exports of sanctioned goods to Russia. We use an event-study design, where the treatment group is composed of sanctioned products (electronics and machines), the control group includes all non-sanctioned products, and February 2022 is the treatment date. We consider the following linear specification:

$$Y_{prt} = \sum_{\tau=21Q1}^{23Q3} \beta_{\tau} \cdot S_p \cdot \mathbf{I}(\tau = q_t) + \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt} \quad (3)$$

The subscript r denotes region, t denotes time (year-month), and p product class. The outcome variable Y_{prt} is exports in million USD, along with the number of transactions and their log values. The binary variable S_p denotes sanctioned products, and q_t is the quarter of month t . Variables κ_r , λ_t , and μ_p are regional, time, and product class fixed effects, respectively. We consider exports to Russia and to other EAEU countries separately.

Figure 5: Value of export growth for sanctioned products: event-study



Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_{τ} . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure 5 shows that after February 2022, exports of sanctioned goods to Russia increased by 10.5M USD per region per month, while exports of these goods to other EAEU countries increased by only around 0.6M USD (Table 2 Columns 1, 2). A similar pattern is observed for

the number of transactions (Figure B1 of Appendix B) – the number of sanctioned product transactions to Russia increased by 483 per region-month, while it is only 39 for other EAEU countries.²⁴ Considering that the production output of electronics and machines has been very stable in Kazakhstan since 2021 (see Figure A5 in Appendix A for raw dynamics and Figure B4 of Appendix B for the event-study) and that a substantial spike in the export of these products to Russia occurred after February 2022 (Figure 2), this result suggests that Kazakh firms started to substantially re-export sanctioned goods to Russia after February 2022.

We now want to understand how the regional language habits affect this re-export activity of sanctioned products – the focus of this section. We consider the following triple DID specification:

$$Y_{prt} = \alpha \cdot S_p \cdot Treat_r + \beta \cdot S_p \cdot Post_t + \gamma \cdot S_p \cdot Post_t \cdot Treat_r + \delta \cdot Border_r \cdot Post_t + \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt} \quad (4)$$

Here, $Treat_r$ is the regional share of Russian-speaking Kazakhs from the 2021 Census (see the right panel of Figure 1) – the treatment exposure. The binary variables are $Post_t = 1$ for periods after February 2022, $Border_r = 1$ for regions with a common border with Russia, $S_p = 1$ for sanctioned products.

The coefficient α measures differences in the export of (not-yet) sanctioned products before February 2022 depending on the treatment exposure. The coefficient β measures the change in export of sanctioned goods after February 2022 for a hypothetical region with zero treatment exposure, while the coefficient γ captures the additional contribution of treatment exposure to this effect. The coefficient δ measures the change in overall exports from regions that share a border with Russia after February 2022. In this specification, our focus is on $\beta + \gamma \cdot Treat_r$ – the marginal effect of (4) by $S_p \cdot Post_t$ for different levels of $Treat_r$, which measures the change in exports of sanctioned products after February 2022 as the share of Russian-speaking Kazakhs varies.

Table 2 presents the results for model (4) for the value of exports to Russia (Column 3) and to other EAEU countries (Column 4) as an outcome variable. The coefficient β is

²⁴Figure B2 shows the similar pattern for the log of the export value and B3 – for the log of transactions

Table 2: The effect of Russian language on sanctions circumvention

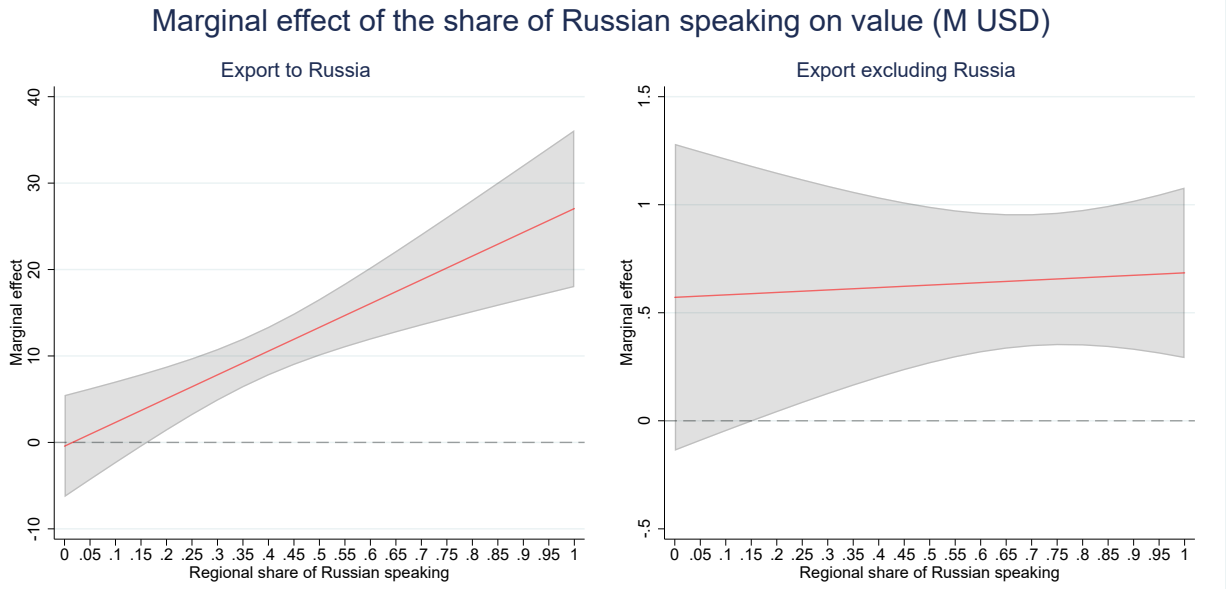
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: export value (in M USD)					
Sanctioned products x treatment exposure (α)			-3.86 (4.35)	0.18 (0.29)		
Sanctioned products x post-invasion (β)	10.5*** (1.45)	0.61*** (0.21)	-0.43 (3.02)	0.57 (0.36)		
Sanctioned products x post-invasion x treatment exposure (γ)			27.5*** (7.07)	0.11 (0.46)	22.4*** (7.13)	-0.69 (0.52)
Border region x post-invasion (δ)			-4.72*** (1.28)	-0.28** (0.11)		
Constant	19.6*** (0.95)	1.64*** (0.065)	20.6*** (1.08)	1.71*** (0.065)	3.17* (1.79)	1.35*** (0.17)
Observations	2,769	2,769	2,769	2,769	537	537
R-squared	0.300	0.435	0.306	0.436	0.524	0.747
Year-Month FE	Y	Y	Y	Y	Y	Y
Region FE	Y	Y	Y	Y	Y	Y
Product FE	Y	Y	Y	Y	N	N
Products	All	All	All	All	Sanctioned	Sanctioned
Sample	All regions	All regions	All regions	All regions	All regions	All regions
Export to	Russia	Non-Russia	Russia	Non-Russia	Russia	Non-Russia

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Columns 1 and 2 show the results of DID estimation – a simplified version of (3). Columns 3 and 4 show the results of a Triple DID estimation of (4), with the regional share of Russian-speaking Kazakh as the treatment exposure. Columns 5 and 6 show the results for (7). Odd columns are for export to Russia, even – for export to other EAEU countries. The dependent variable is the export value (in M USD) for a product class in a region-month. Robust standard errors are in parentheses.

insignificant, suggesting that in a hypothetical region with a zero share of Russian-speaking Kazakhs, the full-scale invasion would have an insignificant effect on the export of sanctioned goods to Russia. The coefficient γ , on the other hand, is positive and significant for exports to Russia, but not for exports to other EAEU countries. It suggests that exports of sanctioned products to Russia after February 2022 increase disproportionately from regions with a high share of Russian-speaking Kazakhs. It implies that the invasion does not move re-exports of sanctioned goods in a region with zero Russian speakers (β insignificant), but the regional prevalence of Russian speakers does ($\gamma > 0$). This is consistent with the notion that supplying Russia is conformity to the interests of the group one identifies with (Prediction 4) rather than a uniform response to a profit opportunity. This becomes even more evident when we plot $\beta + \gamma \cdot Treat$ as a function of $Treat$ (Figure 6) – higher treatment exposure increases exports of sanctioned products to Russia after February 2022, but not to other EAEU countries. Figure B5 of Appendix B shows the estimates of $\beta + \gamma \cdot Treat$ for the number of transactions as

a dependent variable.²⁵ After February 2022, transaction growth increases with treatment exposure for exports to both Russia and other EAEU countries, but the effect is an order of magnitude larger for exports to Russia.

Figure 6: Value of export growth for sanctioned products by regional share of Russian-speaking Kazakh



Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4), where β and γ are taken from Columns 3 and 4 of Table 2. The dependent variable is the value of exports for a product class in a region-month (M USD). The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

To see the dynamics of the treatment exposure and to test for parallel trends, we consider the following event-study analog of (4):

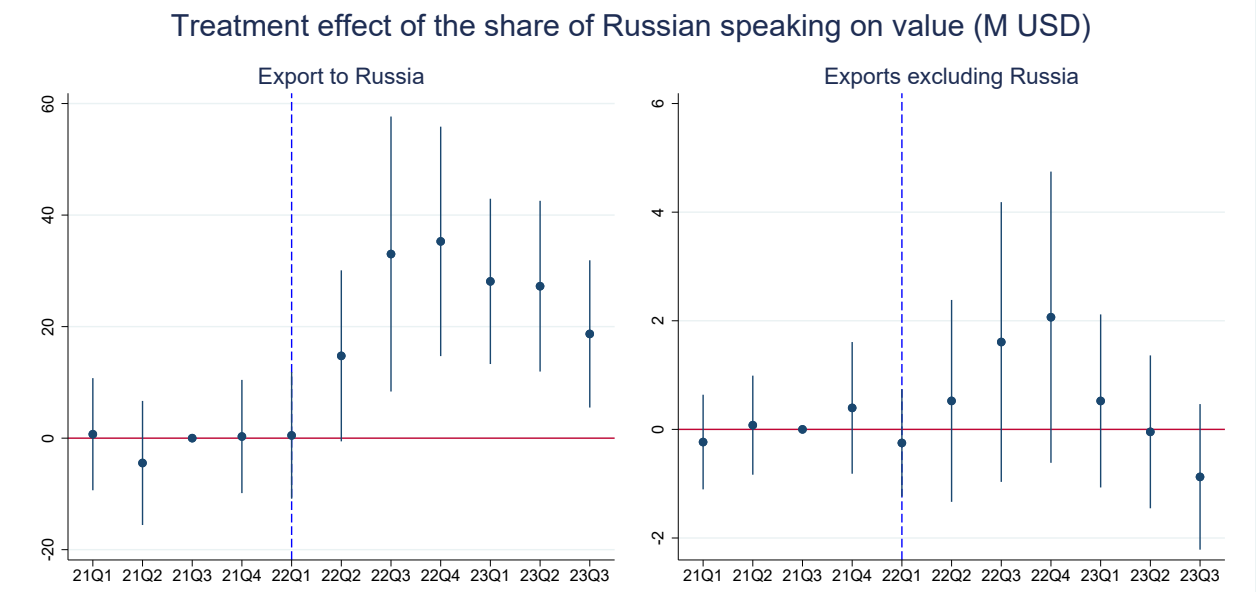
$$Y_{prt} = \alpha \cdot S_p \cdot Treat_r + \beta \cdot S_p \cdot Post_t + \sum_{\tau=21Q1}^{23Q3} \gamma_{\tau} S_p I(\tau = q_t) Treat_r + \delta \cdot Border_r \cdot Post_t + \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt} \quad (5)$$

Figure 7 shows that the parallel trends assumption holds for the export value as a dependent variable. Moreover, the treatment exposure had the strongest effect on the re-export of sanctioned products to Russia in July 2022 - June 2023. Figure B8 shows a similar pattern for

²⁵Figures B6 and B7 show estimates for the log of export value and the log of the transaction number.

the number of transactions as the outcome variable.

Figure 7: Event study for the value of export growth by regional share of Russian-speaking Kazakhstan



Note. The figure shows coefficients γ_τ from (5). The dependent variable is the export value (M USD) for a product class in a region-month.

Finally we also estimate the extension of model (4) to separate the effect of Russian language from the Russian ethnicity:

$$Y_{prt} = \alpha \cdot S_p \cdot Treat_r + \beta \cdot S_p \cdot Post_t + \gamma \cdot S_p \cdot Post_t \cdot Treat_r + \delta \cdot Border_r \cdot Post_t + \eta \cdot S_p \cdot RusEth_r + \theta \cdot S_p \cdot RusEth_r \cdot Post_t + \mu_p + \kappa_r + \lambda_t + \varepsilon_{prt} \quad (6)$$

Here $RusEth_r$ is the share of ethnic Russians in region r . All other variables are similar to equation (4). Figures B9 and B10 show the results for the export value and its log as an outcome variable; Figures B11 and B12 show the results for the number of transactions and its log. The results are similar to those of the baseline model (4), albeit with less statistical power. This suggests that Russian ethnicity does not confound our main results.

These results speak to why export sanctions leak. Most circumvention runs through friendly third countries, which offset roughly two-thirds of the drop in Russia's imports (Egorov et al. 2026; Chupilkin et al. 2026). In this literature, circumvention is arbitrage. Sanctions

make goods scarce in Russia, and firms in neutral neighbors profit from supplying them. Within Kazakhstan all regions face the same Russian demand, and (4) absorbs border logistics through $Border_r \times Post_t$. Profit alone therefore predicts uptake everywhere, with $\beta > 0$ and $\gamma \approx 0$. We find the opposite. A counterfactual region without Russian-speaking Kazakhs would show no re-export growth at all – according to our model. The insignificant α (Table 2, Columns 3–4) adds that Russian-speaking regions had no head start in this trade before the war.

Instead, convictions matter directly. When we re-estimate (4) with the regional share blaming Russia for the war, or not justifying the invasion, as the treatment exposure,²⁶ export growth of sanctioned products declines in this share and turns negative where a majority blames Russia (Figure B13 of Appendix B).²⁷ Section 5.4 develops what it implies for the reading of language as a mere transaction cost.

Finally, we restrict our attention to the subsample of export data, including only sanctioned goods, and estimate a DID specification similar to the one in Korovkin and Makarin (2023):

$$Y_{rt} = \gamma \cdot Post_t \cdot Treat_r + \kappa_r + \lambda_t + \varepsilon_{rt}, \quad (7)$$

with the notation similar to the one in equation (4).²⁸ In this specification, coefficient γ identifies the differential export growth of sanctioned goods after February 2022 between regions with high and low degrees of treatment exposure. Columns 5 and 6 of Table 2 show the results for the value of export to Russia and to other EAEU countries.²⁹ Similar to our previous findings, growth in exports of sanctioned goods to Russia increased from regions with a higher share of Russian-speaking Kazakhs, while it did not for the export to other EAEU countries. An increase in the regional share of Russian-speaking Kazakh from the

²⁶For measuring the regional share of justifying the war and blaming Russia/Ukraine, we use CAB data after February 2022. These shares are measured after the invasion, so we use this pattern not for identification but to tell mechanisms apart.

²⁷Figure B14 shows the mirror image. Export growth of sanctioned products increases with the regional shares blaming Ukraine or the West for the war.

²⁸One can think of regions with low and high treatment exposure as control and treatment groups, respectively, in a binary treatment setting.

²⁹Figures B15 and B16 show the result of the event-study analysis for (7) for the export value and the number of transactions.

minimum (.27) to the maximum (.66) increases additional exports by 8.7M USD per month and region. In line with our previous findings, these results suggest that Russian-speaking Kazakhs significantly increased the re-export of sanctioned products to Russia after the full-scale invasion of Ukraine. Taken together, the trade evidence supports Prediction 4.

5 Mechanisms

We showed that Russian language penetration increases sanctions circumvention. This section establishes the mechanism. We proceed in three steps that mirror the predictions of Section 2. A pre-registered experiment shows that a language cue alone activates identification with Russia and improves attitudes toward it. The CAB panel shows that after the invasion, language polarized people’s attitudes toward Russia. Yandex search data show how media consumption follows the shift and stabilizes it. A final subsection discusses the alternative explanations.

5.1 The Effect of Russian Language on Attitudes to Sanctions

Our difference-in-differences estimates show that Russian-speaking regions re-exported more sanctioned goods to Russia after February 2022. We identify this effect at the regional level, using cross-regional variation in the share of Russian speakers. Even with specifications that control for border regions and Russian ethnicity penetration interacted with the post-invasion period, and despite parallel trends, regional Russian-language penetration is not randomly distributed. In this section, we provide complementary evidence at the individual level, where language itself is randomly assigned, reinforcing that the effect of Russian language identity is causal.

Let binary Z_i indicate that respondent i was randomly addressed in Russian on the intro page, and let R_i indicate that i *completed* the survey in Russian. For an attitude Y_i , our structural equation of interest is

$$Y_i = \beta_0 + \beta_1 R_i + \mathbf{X}_i \delta + \varepsilon_i, \quad (8)$$

where β_1 measures the causal effect of completing the survey in Russian on the attitude, and $\mathbf{X}_i$ is a vector of controls including: ethnicity, gender, age, number of children, education,

occupation, household welfare, internet-use intensity, a rural indicator, and regional fixed effects. Because R_i is a choice and hence potentially endogenous, we instrument it with the randomly assigned language of the introductory page Z_i . Therefore, the first stage is

$$R_i = \pi_0 + \pi_1 Z_i + \mathbf{X}_i \gamma + \nu_i, \quad (9)$$

so that π_1 captures how strongly the assigned language shifts the language respondents actually use. Since Z_i is randomized, we also report the reduced form estimate,

$$Y_i = \alpha_0 + \alpha_1 Z_i + \mathbf{X}_i \theta + u_i, \quad (10)$$

where α_1 is the effect of being nudged into a Russian identity. We report each specification both with and without controls $\mathbf{X}_i$. The specification with controls absorbs residual variation and improves precision. Standard errors are heteroskedasticity-robust.

We begin with the effect of the randomized intro-page language on the language respondents actually use. Table 3 shows the results. Being addressed in Russian raises the probability of *completing* the survey in Russian by about 22 percentage points (Columns 1–2). The randomized intro also increases stated use of Russian with friends and colleagues and for news (Columns 5–8), but not at home (Columns 3–4), where language is most habitual and hardest to move. Completing the survey in Russian is a behavior rather than an answer to a question, so this is our central measure of a Russian-language identity. Moreover, our randomized instrument is a strong predictor of the survey language³⁰, keeping us safely beyond the weak instrument problem (Andrews et al. 2019). The randomly assigned introductory language is also a valid instrument for the survey language by construction of randomization (for balance statistics see Table C2), so IV estimates the compliers’ average causal effect of survey language on attitudes.

Table 4 reports reduced-form regressions of attitudes toward Russian affairs on the randomly assigned intro language. Respondents who receive the Russian intro page are 3 p.p. (12 percent) more likely to consider helping Russia evade sanctions justified (Columns 1–2),

³⁰Kleibergen-Paap Wald rk F-statistic is above 260 in models without controls and above 390 in models with controls (See Table 5).

Table 3: Russian intro page and Russian language use

VARIABLES	(1) Survey in Russian	(2)	(3) Russian at home	(4)	(5) Russian with friends	(6)	(7) News in Russian	(8)
Intro page in Russian	0.23*** (0.014)	0.22*** (0.011)	0.012 (0.014)	0.0022 (0.0091)	0.031** (0.014)	0.023** (0.0097)	0.032** (0.014)	0.023** (0.010)
Constant	0.42*** (0.0098)	0.23*** (0.042)	0.37*** (0.0096)	0.16*** (0.035)	0.40*** (0.0097)	0.079** (0.038)	0.43*** (0.0098)	0.076* (0.041)
Observations	5,025	5,025	5,025	5,025	5,025	5,025	5,025	5,025
R-squared	0.051	0.443	0.000	0.559	0.001	0.519	0.001	0.470
Demographics	N	Y	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y	N	Y
Estimat.	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the effect of randomized intro-page language on Russian-language use across different circumstances. In Columns 1-2, the dependent binary variable equals 1 if the survey was completed in Russian. In Columns 3-4, the dependent binary variable equals 1 if the respondent's primary language for communicating at home is Russian. In Columns 5-6, the dependent binary variable equals 1 if the respondent's primary language for communicating with friends and colleagues is Russian. In Columns 7-8, the dependent binary variable equals 1 if the respondent's primary language for news consumption is Russian. The wording of questions is in Section 3.2. Odd columns do not have controls. Even columns include additional controls: ethnicity, gender, age, number of kids, education, occupation, household welfare, internet use intensity, a rural binary indicator, and regional fixed effects. Robust standard errors are in parentheses.

4 p.p (17 percent) more likely to justify the invasion of Ukraine (Columns 3–4), and 2.7% more favorable opinion of Russia (Columns 5–6). Table 5 reports the resulting IV estimates. Completing the survey in Russian increases justification for helping Russia evade sanctions by 15 p.p. (68%), and for the invasion by 18 p.p. (more than doubles), and raises overall opinion of Russia by 12%. The heteroskedasticity-robust first-stage F-statistic exceeds 20, well above conventional thresholds for strong-instrument inference (Andrews et al. 2019).

According to Prediction 3, the cue activates a Russian identity, which implies that randomly assigned intro pages should activate Russia-related attitudes but not attitudes toward other affairs. Table C3 (reduced form) and Table C4 (IV) in Appendix C show that neither intro page nor survey in Russian have a significant effect on opinions of China or the United States, trust in national government, the justifiability of participating in protests, or overworking.

Finally, a randomized intro page in Russian may affect not only how people think of

Table 4: Reduced form: Russian intro page and attitudes toward Russian affairs

VARIABLES	(1) Sanctions evasion	(2) Sanctions evasion	(3) Invasion/SMO	(4) Invasion/SMO	(5) Opinion of Russia	(6) Opinion of Russia
Intro page in Russian	0.034*** (0.013)	0.030** (0.013)	0.041*** (0.012)	0.039*** (0.012)	0.091*** (0.031)	0.080*** (0.029)
Constant	0.28*** (0.0089)	0.47*** (0.051)	0.24*** (0.0084)	0.45*** (0.047)	3.32*** (0.022)	3.78*** (0.11)
Observations	5,025	5,025	5,025	5,025	4,334	4,334
R-squared	0.001	0.072	0.002	0.105	0.002	0.137
Demographics	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Estimat.	OLS	OLS	OLS	OLS	OLS	OLS

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the effect of randomized intro-page language on attitudes toward Russian affairs. In Columns 1-2, the dependent binary variable equals 1 if helping Russia in sanctions evasion is “justified”. In Columns 3-4, the dependent binary variable equals 1 if the invasion of Ukraine/Special military operation is “justified”. In Columns 5-6, the dependent scale variable is the opinion of Russia (from 1 – “Very negative” to 5 – “Very positive”). The wording of questions is in Section 3.2. Odd columns do not have controls. Even columns include additional controls: ethnicity, gender, age, number of kids, education, occupation, household welfare, internet use intensity, a rural binary indicator, and regional fixed effects. Robust standard errors are in parentheses. Table C3 shows results for attitudes toward non-Russian affairs.

Russia but also their perception of how prevalent Russian is in their daily lives. We have already shown the reduced-form results in Table 3 for the Russian language use in different circumstances. Table C5 in Appendix C shows IV results, showing that completing the survey in Russian raises respondents’ reported use of Russian with friends and for news. A short, random exposure to Russian thus shifts both behavior and stated language use, consistent with the view that the cue activates an identity. That the effect appears for language used with friends and for news, but not at home, fits this reading. The cue activates the more public, identity-expressive domains of language use, while the most habitual domain is slower to move. Taken together, these results support Prediction 3 and establish the causal relation. The effect is specific to Russian affairs, expressed in revealed behavior as well as in stated preferences. It operates even in the absence of any media exposure, identifying a standalone role for language. These attitudes, in turn, induce the re-export of sanctioned goods documented in Section 4.

Table 5: The effect of Russian survey language on attitudes toward Russian affairs

VARIABLES	(1) Sanctions evasion	(2) Sanctions evasion	(3) Invasion/SMO	(4) Invasion/SMO	(5) Opinion of Russia	(6) Opinion of Russia
Survey in Russian	0.15** (0.059)	0.14** (0.059)	0.18*** (0.057)	0.18*** (0.055)	0.38*** (0.13)	0.34*** (0.12)
Constant	0.22*** (0.032)	0.44*** (0.054)	0.16*** (0.031)	0.41*** (0.052)	3.17*** (0.068)	3.70*** (0.12)
Observations	5,025	5,025	5,025	5,025	4,334	4,334
Demographics	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Estimat.	IV	IV	IV	IV	IV	IV
Kleibergen-Paap Wald rk F	269.3	408.7	269.3	408.7	266.4	391.5

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the IV results of the effect of completing the survey in Russian on attitudes toward Russian affairs. The dependent variables (and control variables in the even columns) are the same as in Table 4. The heteroskedasticity-robust F-statistic shows that the instrument is strong (≥ 20) for standard inference (Andrews et al. 2019). Robust standard errors are in parentheses. Table C4 shows results for attitudes toward non-Russian affairs. Table C5 shows results for Russian language use across different circumstances.

5.2 The Polarizing Effect of the Russian Language

So far, we have shown that Russian-language identity increases pro-Kremlin views in Fall 2023. The relevant question is whether Russian speakers have always been more pro-Russian than non-Russian speakers, as they could share long-term historical and cultural ties with Russia, or whether the full-scale invasion triggered an identity affiliation. The CAB data allow us to test Predictions 1 and 2 together. The war should lower the average opinion of Russia for everyone, because the status shock is common to everyone (decrease of S_R). At the same time, it should open a gap by language, because the salience shock magnifies differences that already exist. To examine this argument, we use questions on attitudes to Russia, the US, and China from eight CAB waves from 2020 to 2023, allowing us to estimate the following DID specification:

$$y_{irt} = \alpha Treat_r + \beta Post_t + \gamma Treat_r \cdot Post_t + \mathbf{X}_{irt} \delta + \epsilon_{irt} \quad (11)$$

Here, i denotes a respondent from region r and survey wave t . The outcome variable y_{irt} is the opinion of Russia/US/China on a 4-level Likert scale (1 - 'very unfavorable', 4 - 'very favorable'), $Treat_r$ is the regional share of Russian-speaking Kazakhs from Census 2021, and $Post_t$ is the binary indicator for waves after February 2022. The vector $\mathbf{X}_{irt}$ includes the following set of control variables: gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer's assessment of the respondent's survey comprehension and comfort with answering the survey. The control for individual ethnicity enables us to separate ethnic identity from the regional-level language effect.

The coefficient α captures the language gap in y_{irt} before February 2022, depending on the regional share of Russian-speaking Kazakhs. Coefficient β shows the trend in y_{irt} after February 2022 for a hypothetical region with zero Russian-speaking Kazakhs. Under the parallel trends assumption, the coefficient γ measures divergence from the trend, depending on the share of Russian-speaking Kazakhs after February 2022, i.e. the effect of the Russian language penetration on y_{irt} after February 2022.

We also consider the extension of (11) by introducing the region (κ_r) and survey wave (λ_t) fixed effects, to take into account all the variation (between regions and over time) that is not related to the Russian language penetration and the full-scale invasion.

$$y_{irt} = \gamma Treat_r \cdot Post_t + \mathbf{X}_{irt}\delta + \kappa_r + \lambda_t + \epsilon_{irt} \quad (12)$$

Table 6 shows the estimates for (11) (Columns 1,3,5) and (12) (Columns 2,4,6). Coefficient α is insignificant for the opinions of Russia and the US, and positive for China. This suggests that before the full-scale invasion, respondents from regions with a higher share of Russian-speaking Kazakhs did not have more positive views of Russia and the US, but did have more positive views of China. Coefficient β is negative for Russia, insignificant for the US, and positive for China, suggesting that the view of Russia (China) substantially deteriorated (improved) in regions with a low share of Russian-speaking Kazakhs. This negative and Russia-specific β corresponds to Prediction 1. The war lowers Russia's status for all residents, while opinions of the US do not fall. Finally, Russian-language penetration positively affected

only the opinion of Russia after February 2022 (γ), with the opposite sign to β , suggesting polarization of opinions about Russia after February 2022 between regions with high and low shares of Russian-speaking Kazakhs. The positive γ works against this common decline and corresponds to the widening gap of Prediction 2. The insignificant α confirms the second part of Prediction 2. Before the war, language carried little attention weight and predicted opinions of Russia only modestly. Because the treatment is the share of Russian-speaking *Kazakhs* and we control for individual ethnicity, the gap opens along language rather than ethnicity, as Prediction 2 suggests.

Figure 8 shows event-study coefficients for the DID specification (12). For all three outcome variables, there are parallel trends before February 2022, and only 'Opinion of Russia' is positively affected by the regional share of Russian-speaking Kazakhs after February 2022.³¹ The gap also does not fade over the sample period, as the event-study coefficients remain elevated through late 2023. Prediction 5 attributes this persistence to the media feedback that we document in Section 5.3.

Overall, the CAB evidence supports Predictions 1 and 2. The war lowered everyone's opinion of Russia and at the same time polarized regions along language. The argument that Russian speakers have simply always been closer to the Russian world is therefore not supported. In the next section we turn to the media demand channel of Prediction 5. Those who newly identify with R should demand more Russian-language media, which in turn feeds the polarization back.

5.3 The Effect of Russian Language on Media Choice

So far, we have shown that the Russian language itself increases justification of the invasion, sanctions circumvention, and opinion of Russia. We also demonstrated that this gap in pro-Russian opinion emerged only after February 2022. This section tests the first component of Prediction 5. Demand for Russian-language media should rise after February 2022 as a further

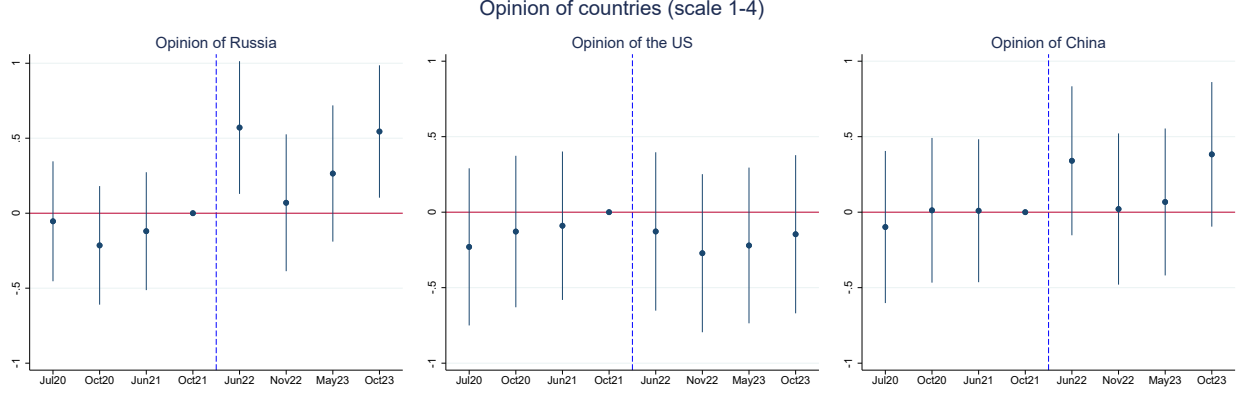
³¹Table D14 and Figure D4 show the DID and event-study estimates for the two language outcome variables: 'Russian at home' and 'Survey in Russian'. The Census 2021 regional share of Russian-speaking Kazakhs does not affect the probability of using the Russian language at home after February 2022, while it increases the probability of completing surveys in Russian after February 2022, reinforcing the polarization argument rather than consolidation, as was shown for Ukraine (Abramenko et al. 2026).

Table 6: Opinion of Russia, US and China

VARIABLES	(1) Opinion of Russia	(2) Opinion of Russia	(3) Opinion of the US	(4) Opinion of the US	(5) Opinion of China	(6) Opinion of China
Treatment exposure (α)	-0.12* (0.069)		0.052 (0.087)		0.48*** (0.084)	
Post-invasion (β)	-0.57*** (0.047)		-0.020 (0.054)		0.14*** (0.051)	
Treatment exposure x Post-invasion (γ)	0.44*** (0.11)	0.47*** (0.11)	-0.073 (0.13)	-0.079 (0.13)	0.20* (0.12)	0.22* (0.12)
Constant	3.22*** (0.040)	3.12*** (0.099)	2.67*** (0.050)	2.68*** (0.11)	2.46*** (0.047)	2.76*** (0.10)
Observations	11,934	11,934	10,759	10,759	11,497	11,497
R-squared	0.122	0.137	0.111	0.117	0.120	0.131
Time FE	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Mean DV	2.808	2.808	2.625	2.625	2.498	2.498

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The odd columns show results of regression (11), even columns – (12) for scaled dependent variables: “Opinion of Russia” (columns 1-2), “Opinion of the US” (columns 3-4), “Opinion of China” (columns 5-6). The treatment exposure is the regional share of Russian-speaking Kazakhs from Census 2021. The set of control variables includes gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey.

Figure 8: Event-study for Opinions of Russia, the US, and China by the regional share of Russian-speaking Kazakh.



Note. The figure shows event-study coefficients γ_t for the DID specification (12) and dependent variables: “Opinion of Russia” (left), “Opinion of the US” (middle), “Opinion of China” (right).

expression of conformity by those who newly identify with the Russian-speaking world.

To test this hypothesis, we use Yandex search statistics at the month-by-region level in this section. To see whether Russian-language penetration affected the frequency of searches for war-related information, we use the Census 2021 share of Russian-speaking Kazakhs at the regional level and match it to regional search statistics from Yandex. We consider the following event-study specification:

$$y_{rt} = \sum_{\tau} \gamma_{\tau} \mathbf{I}(\tau = t) Treat_r + \kappa_r + \lambda_t + \epsilon_{rt} \quad (13)$$

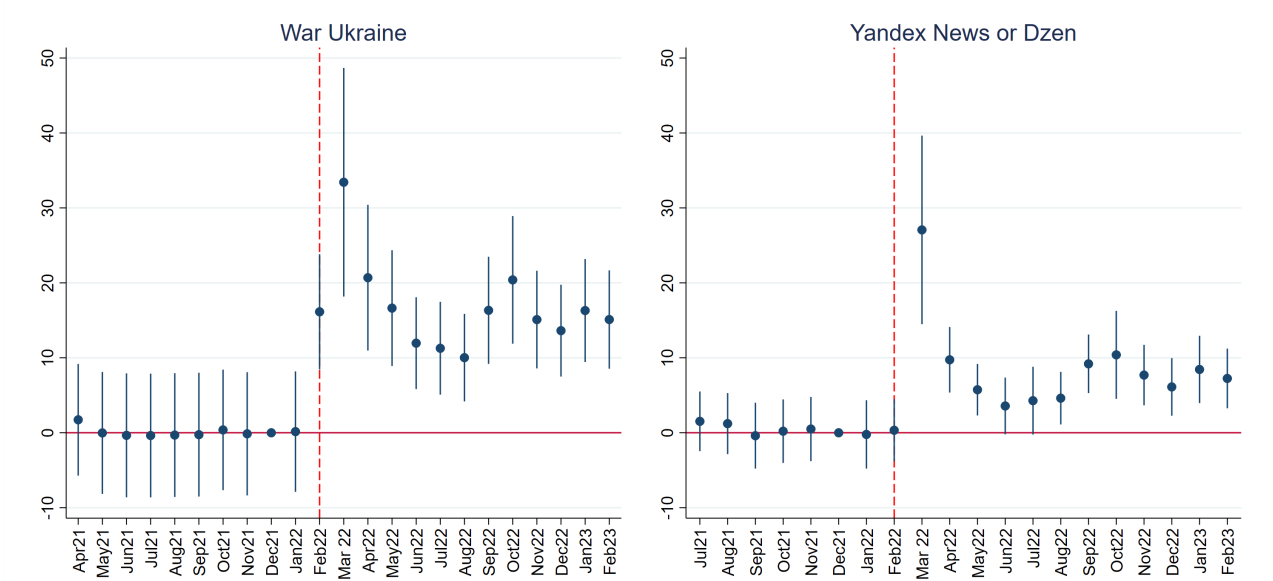
The dependent variable $y_{rt} = \frac{\text{Num. of searches in region } r \text{ in month } t}{\text{Population in region } r} \cdot 1000$ is the number of searches for a specific keyword per month-region for 1000 inhabitants. $Treat_r$ is the share of Russian-speaking Kazakhs in region r according to the Census 2021 (Figure 1), κ_r is the region fixed effect, and λ_t is the year-month fixed effect. The coefficient γ_{τ} captures the differential effect of regional Russian-language penetration on the frequency of online searches in month t .

The left graph of Figure 9 shows the results for the keywords “vojna Ukraina” (“war Ukraine”, see also Figure 4).³² It shows that regions with a higher share of Russian speakers experienced a significant increase in searches for war-related keywords compared to regions

³²The Yandex search statistics are insensitive to letter case, prefixes, suffixes, pronouns, and prepositions, as a search phrase is first lemmatized and stopwords are removed.

with a lower share of Russian speakers, following the February 2022 event. Specifically, increasing the share of Russian speakers from 27% to 66% in a region (that is, from the lowest to the highest share in Kazakhstan) increases the number of searches for war-related keywords by 13.3 per 1,000 people.³³ This language gap spike in search for war-related information in Russian does not mean that people automatically ended up on pro-Kremlin websites. To investigate this further, we examine the search patterns for specific news aggregators and outlets.

Figure 9: Event-study coefficients for Yandex search “war Ukraine”, “Yandex news” or “Yandex Dzen”



Note. The figure shows coefficients γ_τ from (13) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1000 regional inhabitants.

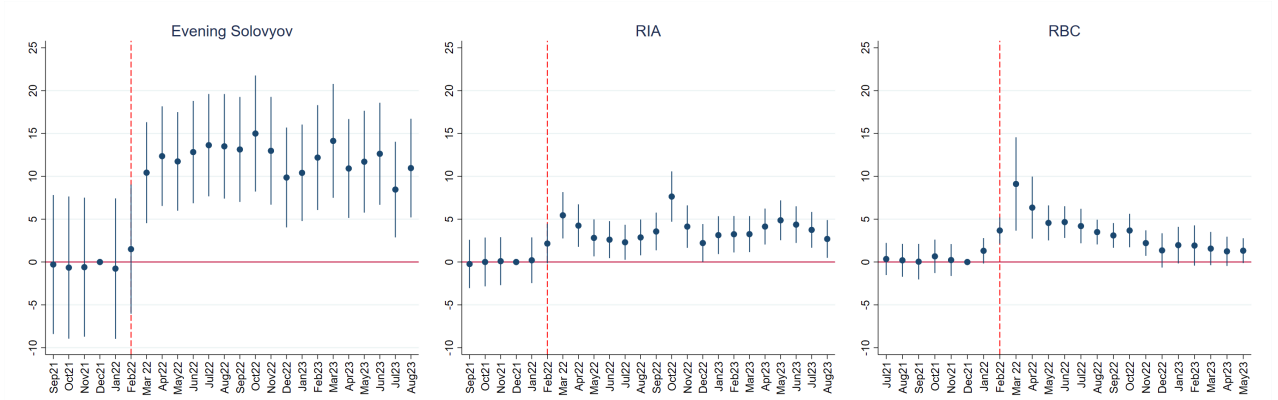
First, we estimate equation (13) for searches for Yandex Dzen or Yandex News – the largest pro-Kremlin news aggregator.³⁴ The right graph of Figure 9 shows that the effect of the regional share of Russian speakers on searches for Yandex News is comparable to that

³³Figure E2 in Appendix E shows that the large-scale protests that occurred in Kazakhstan in January 2022 – the most significant protests that ever happened in Kazakhstan’s history as an independent country – led to a much smaller language gap in the search for information, highlighting that the interest of Russian speaking population in search for information induced by Russia’s full-scale invasion of Ukraine was truly exceptional.

³⁴Yandex News was acquired by the Russian state-owned social media VK in 2022, and renamed Yandex Dzen.

for “war Ukraine”. This suggests that Yandex News is an important source of news in the Russian-speaking regions of Kazakhstan after February 2022. Figure 10 shows that other pro-Kremlin TV shows and media outlets, such as “Evening with Solovyov”, RIA News, and RBC News, also experienced a significant surge in Russian-speaking regions after February 2022.

Figure 10: Event study design for Yandex search of pro-Kremlin news outlets



Note. The figure shows coefficients γ_τ from (13) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1000 regional inhabitants.

Searches for Russian-language versions of the BBC and Euronews (the most popular pro-Western outlets in Kazakhstan), on the other hand, showed only a very small spike in Russian-speaking regions, suggesting that Western outlets were unable to compete in Kazakhstan’s Russian-language information space (Figure E1). Finally, Figure E3 shows that the effect of Russian-language penetration on news searches after February 2022 is specific to the war in Ukraine, and that other generic searches, such as weather, online games, or movies, are not connected to this event. All in all, the search data confirm the media demand component of Prediction 5. Russian-language penetration sharply increases searches for pro-Kremlin news after February 2022, and specifically for war-related content.

5.4 Discussion

Each of our data sources addresses a distinct threat to a causal reading of the others. The trade data show that Russian language penetration increases the re-export of sanctioned products to Russia; the survey experiment highlights the exclusive effect of the language

identity; the sequence of the survey data shows the language-based polarization appears only after February 2022, and the online search data highlights media as one of the mechanisms. Together they support a simple reading. The war raised the salience of the linguistic divide, salience strengthened identification with the Russian world, and this identification moved both stated attitudes and the re-export of sanctioned goods. The rest of this subsection asks how far alternative readings carry, and what they leave for identity to explain.

The first alternative reading is profit, and it is partly right. Sanctions made goods scarce in Russia, and in the rerouting literature circumvention is arbitrage, so firms from anywhere in the customs union should collect the rents (Egorov et al. 2026). Our aggregate estimates show exactly this rent-driven rerouting. What arbitrage alone does not predict is the gradient, since a counterfactual region without Russian-speaking Kazakhs would have seen no re-export growth at all (Section 4). Language plausibly does lower the cost of finding and contracting with Russian buyers (Melitz 2008; Melitz and Toubal 2014). This variant predicts the same trade gradient as identity, so the trade data alone cannot reject it. The attitude evidence from Section 5.1 shows what transaction costs alone leave unexplained. A contracting advantage implies no reaction to a randomized intro page, yet the cue raises stated justification of sanctions evasion by 3 percentage points. It implies no break in February 2022, since the war changed nothing about who speaks Russian, yet the opinion gap opens exactly then. And it implies that exports follow language rather than conviction, yet export growth falls with the regional share blaming Russia for the war and turns negative where a majority does (Section 4). A last variant runs the causality backward, with the re-export boom producing the attitudes that justify it. But the randomized cue moves attitudes with no transaction involved, and the attitude gap opens among respondents who do no trading, in the first CAB wave after the invasion and before the surge peaks in late 2022. We conclude that opinions moved first and trade followed, matching the broader pattern that citizens adjust their attitudes to their identities rather than the reverse (Pop-Eleches and Robertson 2018). Profit explains why circumvention happens but identity predicts who engages in it.

A skeptic might instead read the survey responses as cheap talk. Survey answers always carry an element of self-presentation, and part of the cue effect may well be conformity. Even so, three observations remain. First, the cue does not make respondents more agreeable in

general, since opinions of China and the United States, trust in the national government, and the work-hours placebo stay flat. Second, presenting oneself as part of the Russian-speaking world is itself an identity choice and choosing whose approval to seek is exactly the conformity behavior the framework models (Shayo 2020; Jia and Persson 2021). Third, the trade data back the stated answers with behavior. The informative behavior is not the trading itself, because re-export was profitable and profit alone explains why someone takes a profitable deal. What profit cannot explain is turning one down. In regions where majorities blame Russia for the war, export growth of sanctioned goods turns negative, so these regions leave real money uncollected (Section 4). Forgoing available profits is the costly counterpart of the stated attitudes, the kind of trade-off that separates identity from talk (Atkin et al. 2021).

A final alternative attributes the attitude shift to persuasion by Russian media rather than to identity. Section 5.3 documents how strongly Russian-speaking regions turned to pro-Kremlin outlets after the invasion. In our setting, persuasion cannot be the activating force, because the experimental cue moves attitudes with no media exposure. We therefore place media downstream of identity. Language decides which identity the war activates, the activated identity decides which media people seek out, and the media then amplify and stabilize the shift. Experimental work on foreign media points the same way, since exposure to a foreign power’s outlet persuades those already close to it and repels the rest, deepening the divide it meets rather than creating a new one (Kao 2026). When outlets of opposite slant both deepen the divide they meet, the direction of the effect comes from the receiver’s identity and not from the content (Bisgaard and Slothuus 2018; Slothuus and Bisgaard 2021). This heterogeneity is what the psychology of motivated reasoning predicts. People accept messages that fit their prior with little scrutiny and argue against messages that do not, so identical information can push groups with opposite priors apart (Kunda 1990; Taber and Lodge 2006). The divergence in the CAB panel is the field version of this pattern. Both language groups faced the same war and the same facts, and their beliefs about it moved apart. Identity supplies the direction of the motive, and the war decided which identity does. The same experiments suggest why the shift then persists. Inside a like-minded online network, news that contradicts the group does not change minds, because the network supplies the counterarguments (Enikolopov et al. 2026). This polarization is a result of *confirmation*

bias – the tendency to interpret the world congruent with beliefs (Nickerson 1998) – the key psychological mechanism that explains both motivated reasoning (Taber and Lodge 2006; Epley and Gilovich 2016) and group based conformity in social identity theory (Tajfel and Turner 1986). We argue that is also affected citizens in Kazakhstan the same way as it did US citizens in the last two decades (Druckman et al. 2013), UK citizens after the Brexit (Hobolt and Tilley 2026), or a global population along the lines of vaccination uptake (Gollwitzer et al. 2020).

6 Conclusion

Kazakhstan watched Russia’s war from outside, yet its citizens did not watch it in the same way. Where Russian is the language of daily life, people moved toward Russia in their attitudes toward the war, in the language they report living in, and in the sanctioned goods their regions ship across the border. Our evidence traces these movements to one mechanism. The invasion made language salient as a marker of identity, the activated identity shifted attitudes toward Russia, and the attitudes carried into trade.

The bystander’s response speaks to when war unites a country and when it divides one. In attacked Ukraine, the invasion made the boundary between the two states salient, a trait all residents share, and the country consolidated (Abramenko et al. 2026). Nation building of this kind requires that the war strike the nation as a whole (Sambanis et al. 2015). Kazakhstan was not struck, so the war instead raised the weight on a trait that divides its residents, and the country polarized. Consolidation and polarization are one mechanism under different patterns of exposure rather than competing theories.

The trade findings carry the same logic into economics. In the attacked country, conflict severs trade between groups because it destroys the trust their exchange rests on (Korovkin and Makarin 2023). In the bystander, a shared language keeps identification with the other side intact, and the identification sustains the exchange. Conflict does not simply reduce intergroup trade. It reallocates trade along the boundaries of identity, cutting exchange where identification breaks and preserving it where identification survives.

For sanctions policy this demand side matters. Export controls target firms, products, and routes. The supply of willing intermediaries, however, follows identity, a margin that export

controls are not built to reach. Moving from the least to the most Russian-speaking region of Kazakhstan raises the re-export of sanctioned goods by 8.7 million USD per region and month, a flow organized by loyalty as much as by logistics.

The polarization also has a beneficiary. In Russia itself, media that polarize can serve the very power they divide people over, because mobilizing supporters outweighs losing opponents (Enikolopov et al. 2026). Our results carry this logic of polarize and rule across a border. Russia did not need to persuade the median citizen of Kazakhstan. Activating the identification of its Russian-speaking neighbors was enough, and the return came not in votes but in sanctioned goods.

The mechanism travels beyond Kazakhstan. Wherever the language of a former metropole still sorts citizens into identities, a war fought in that language’s name can polarize a country that is not itself at war. The policy levers of such countries are slow, since language use changes over generations, as Kazakhstan’s gradual shift to the Latin alphabet illustrates. What our design cannot answer is why the war activated the linguistic boundary inside the country rather than uniting the country against the aggressor next door. When a bystander consolidates and when it polarizes remains an open question for research on the wars that empires leave behind.

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Online Appendix

A Appendix

Example for the conceptual model. Take three binary traits and set $\delta_k = 0$ when i matches the group’s prototype on trait k and $\delta_k = 1$ when she does not.

The attacked country. In Ukraine, the war makes a trait salient that carried little weight before, namely the *side* of the interstate boundary a person stands on. Suppose the prototype of R is (ethnic Russian, Russian-speaking, Russia’s side) and that of the Ukrainian nation N is (ethnic Ukrainian, Ukrainian-speaking, Ukraine’s side). Let us consider a Russian-speaking ethnic Russian resident of Kharkiv who matches R on the first two traits but stands on Ukraine’s side. Before the invasion, with equal weights, she is

$$d_{iR} = \frac{1}{3}(0) + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{1}{3}, \quad d_{iN} = \frac{1}{3}(1) + \frac{1}{3}(1) + \frac{1}{3}(0) = \frac{2}{3},$$

closer to R . When the invasion raises the weight on the interstate side to $\frac{3}{5}$ (leaving $\frac{1}{5}$ on each other trait),

$$d_{iR} = \frac{1}{5}(0) + \frac{1}{5}(0) + \frac{3}{5}(1) = \frac{3}{5}, \quad d_{iN} = \frac{1}{5}(1) + \frac{1}{5}(1) + \frac{3}{5}(0) = \frac{2}{5},$$

she flips to N . A Ukrainian-speaking ethnic Ukrainian sits at $d_{iN} = 0$ and $d_{iR} = 1$ under any weights and does not move. The reweighting therefore shifts only the Russian-leaning residents, and shifts them toward N – consolidation, largest exactly where pre-war attachment to N was weakest, as [Abramenko et al. \(2026\)](#) find.

The bystander. Kazakhstan is not a party to the war, so no “side” trait exists for its residents; the relevant traits are language, ethnicity, and culture. Suppose the prototype of the Russian-speaking world R is (ethnic Russian, Russian-speaking, Orthodox) and that of the Kazakh nation N is (ethnic Kazakh, Kazakh-speaking, Muslim). A person i who is ethnic Kazakh, Russian-speaking, and atheist, weighting the three traits equally ($w_k = 1/3$), is then

$$d_{iR} = \frac{1}{3}(1) + \frac{1}{3}(0) + \frac{1}{3}(1) = \frac{2}{3}, \quad d_{iN} = \frac{1}{3}(0) + \frac{1}{3}(1) + \frac{1}{3}(1) = \frac{2}{3},$$

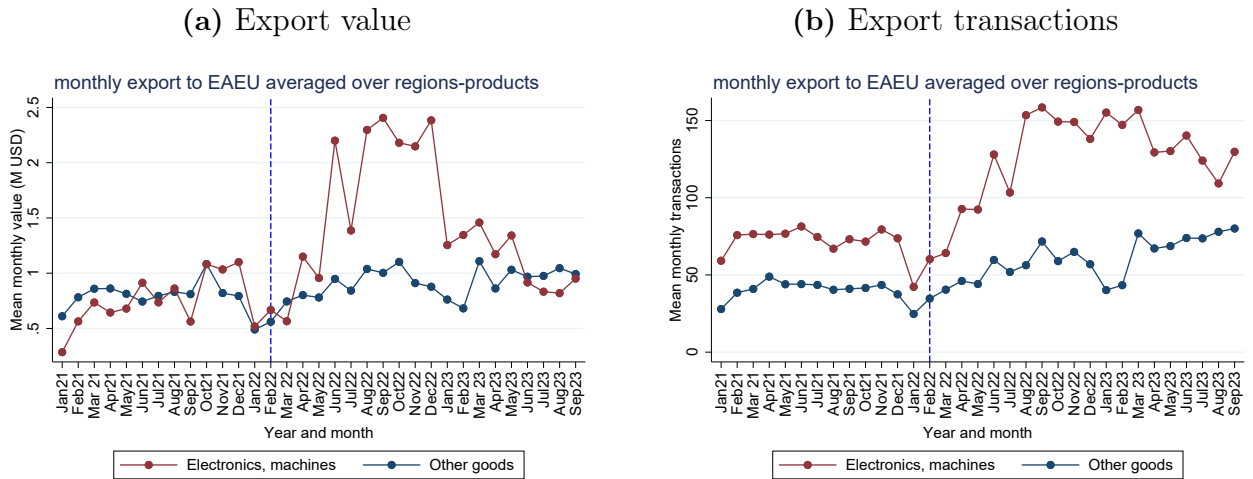
equally distant from both prototypes. Raising the weight on language alone to $1/2$ pulls her toward the group whose language she shares:

$$d_{iR} = \frac{1}{4}(1) + \frac{1}{2}(0) + \frac{1}{4}(1) = \frac{1}{2}, \quad d_{iN} = \frac{1}{4}(0) + \frac{1}{2}(1) + \frac{1}{4}(1) = \frac{3}{4},$$

so she now sits closer to R than to N , while her Kazakh-speaking neighbor with other similar traits stays closer to N .

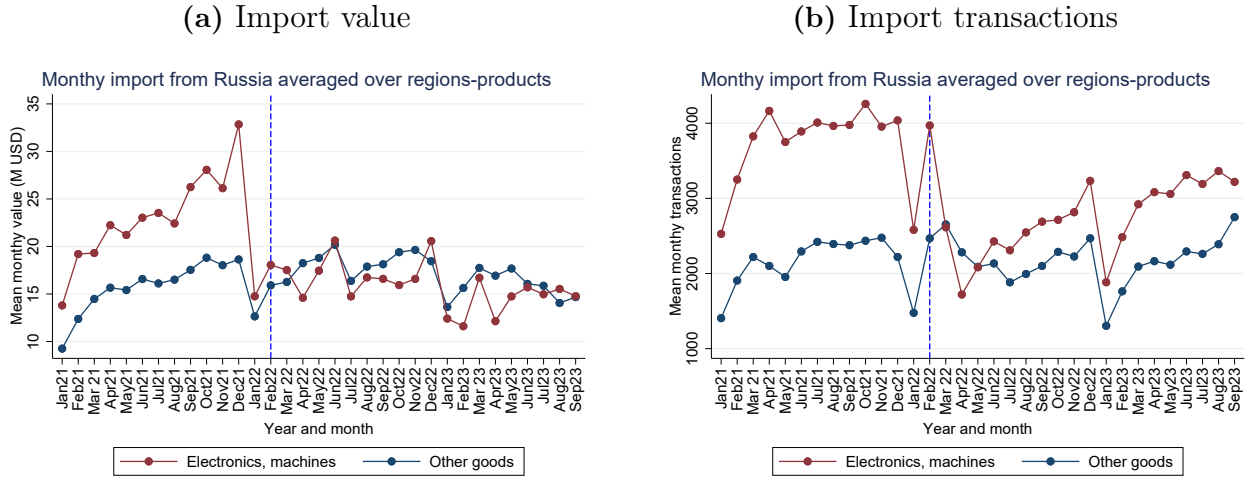
The two cases differ in one structural feature. In the attacked country the war lifts the weight on a trait every resident shares (being on the attacked side), so the reweighting pulls all residents toward N together. In the bystander the war lifts the weight on a trait residents sit on opposite ends of (language), so the same mechanism pulls them apart. Whether war consolidates or polarizes thus depends on whether the trait it makes salient unites or divides the population.

Figure A1: Export from Kazakhstan to EAEU countries excluding Russia



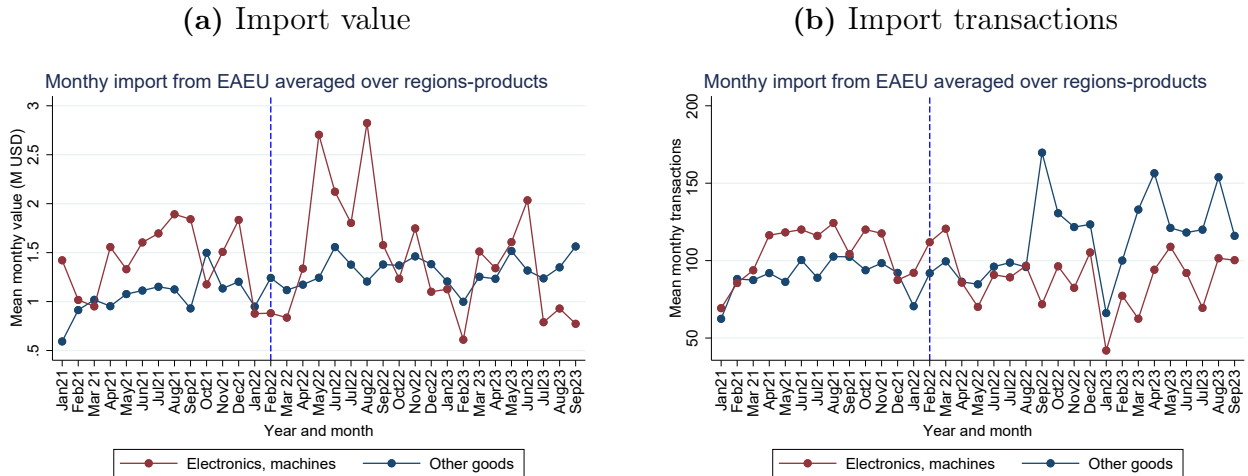
Note: the figure shows the monthly value of export in millions USD (left) and the number of export transactions (right) of Kazakhstan to EAEU countries excluding Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Figure A2: Import to Kazakhstan from Russia



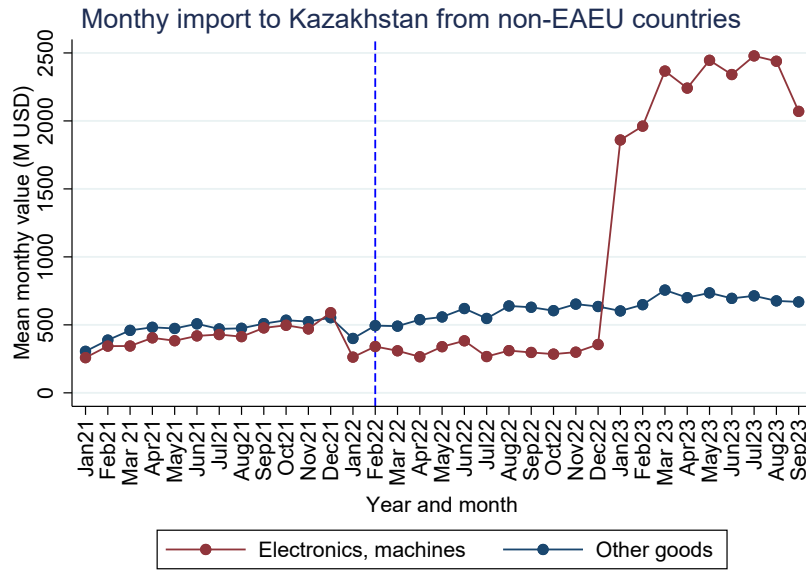
Note: the figure shows the monthly value of imports in millions USD (left) and the number of import transactions (right) to Kazakhstan from Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Figure A3: Import to Kazakhstan from EAEU countries excluding Russia



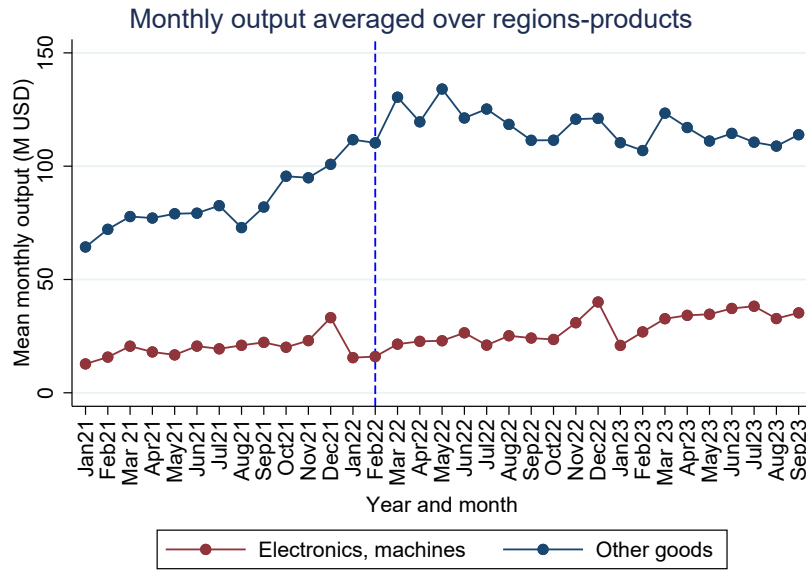
Note: the figure shows the monthly value of imports in millions USD (left) and the number of import transactions (right) to Kazakhstan from EAEU countries excluding Russia, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Figure A4: World import to Kazakhstan



Note: the figure shows the monthly world value of imports to Kazakhstan in millions of USD, averaged across 5 product classes for 2021-2023. The data are collected from the [UN Comtrade Database](#) at the monthly-HS2 code level, covering total imports into Kazakhstan from the world.

Figure A5: The production output of Kazakhstan



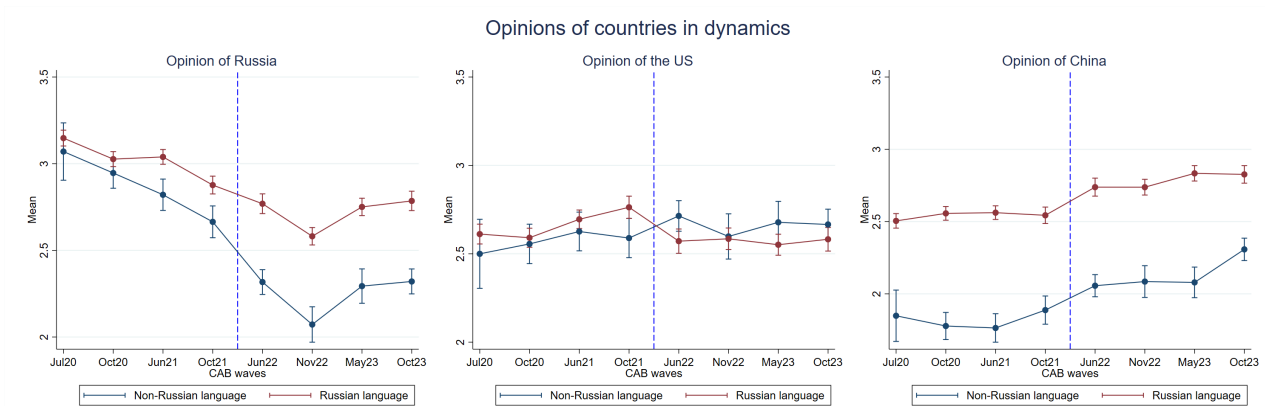
Note: the figure shows the monthly value of production, averaged over 5 product classes and regions for 2021-2023. For more details, see Section 3.

Table A1: Descriptive statistics for the war attitudes and responsibility attribution

Year	SMO is justified	War responsibility (%)		
		Russia	West	Ukraine
2022	26.3	27.5	10.8	20.8
2023	25	24.9	12.1	19.4

Note. The table uses CAB survey waves for 2022-2023. The “SMO is justified” column shows the percentage of “Completely justified” or “Somewhat justified” opinions.

Figure A6: Opinions of Russia, the US and China by survey language over time



Note. The figure shows the mean trends in opinions across different countries by survey language (Russian or Kazakh), based on CAB data for 2020-2023.

Figure A7: Search for “War Ukraine” in Kazakh language on Yandex

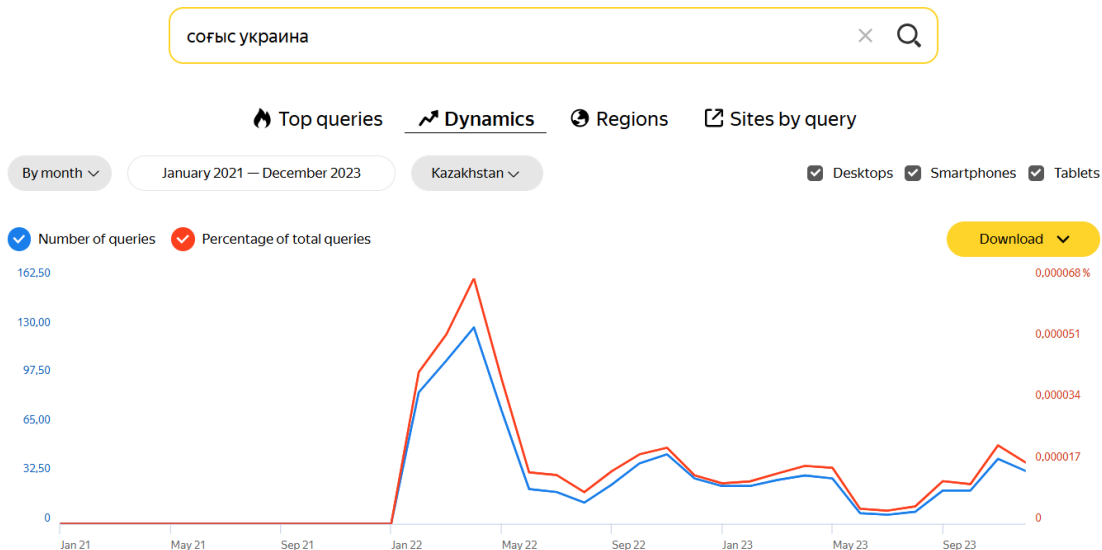


Figure A8: Search for “War Ukraine” in Kazakh language on Google Trends

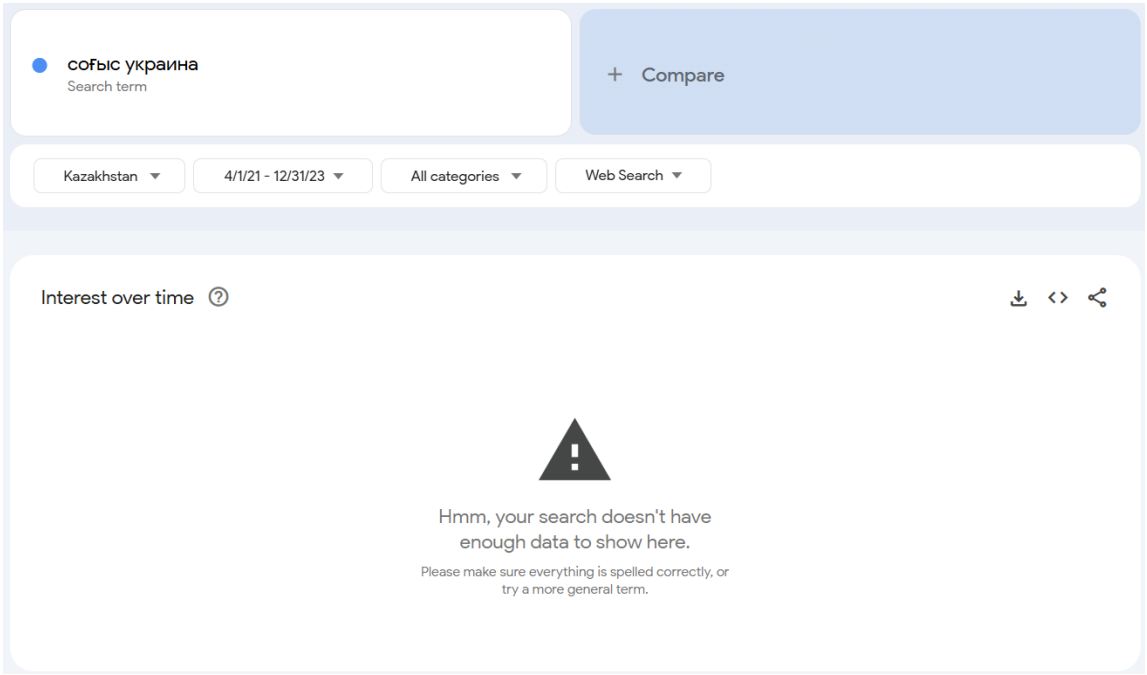


Figure A9: Search for “News” on Yandex

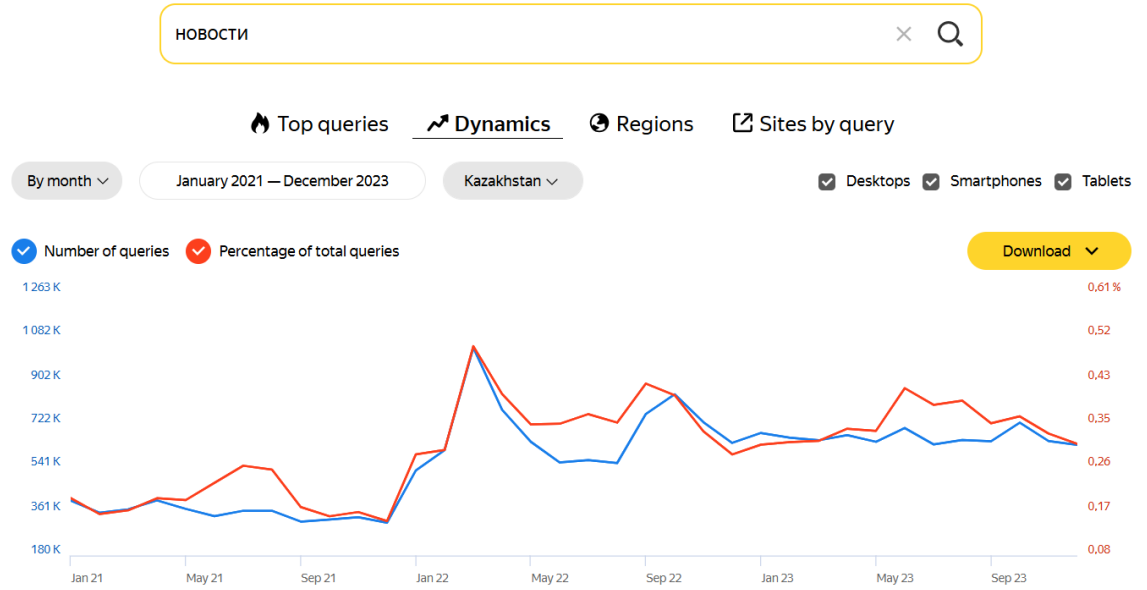


Figure A10: Search for “Yandex Dzen” on Yandex

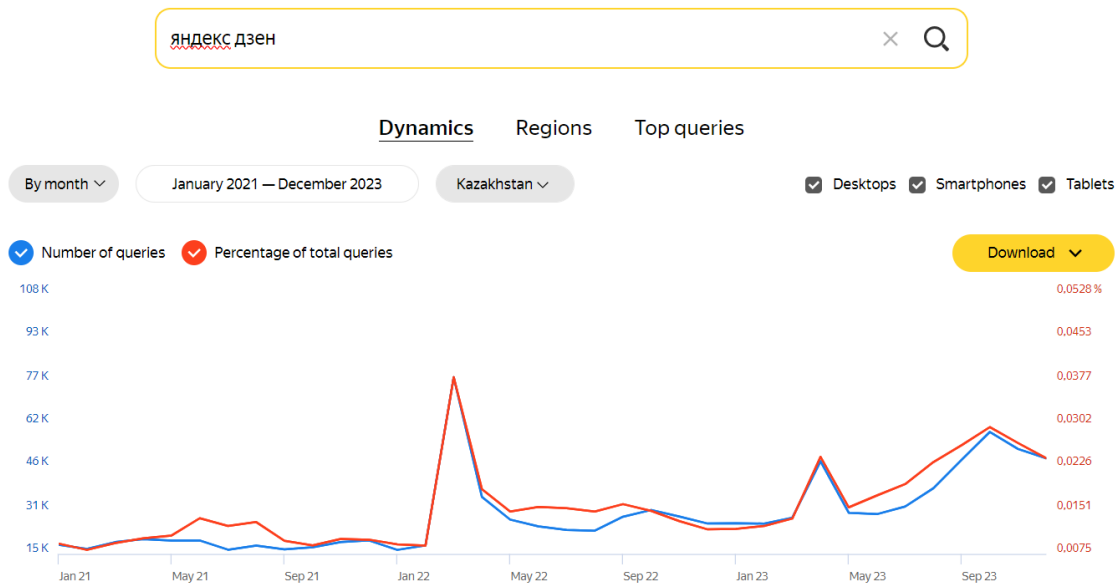


Figure A11: Search for “Yandex News” on Yandex

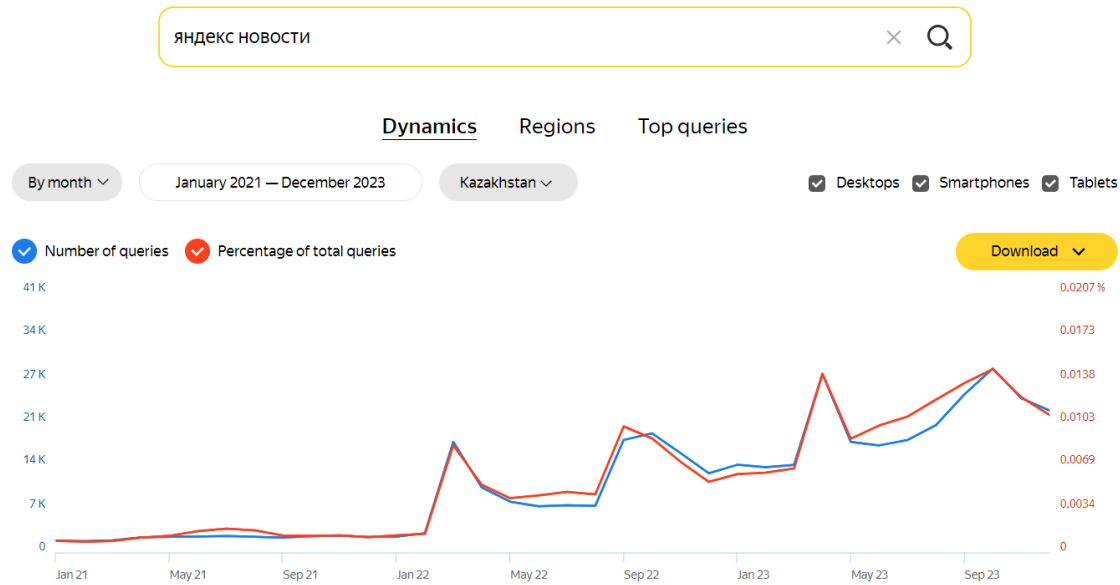
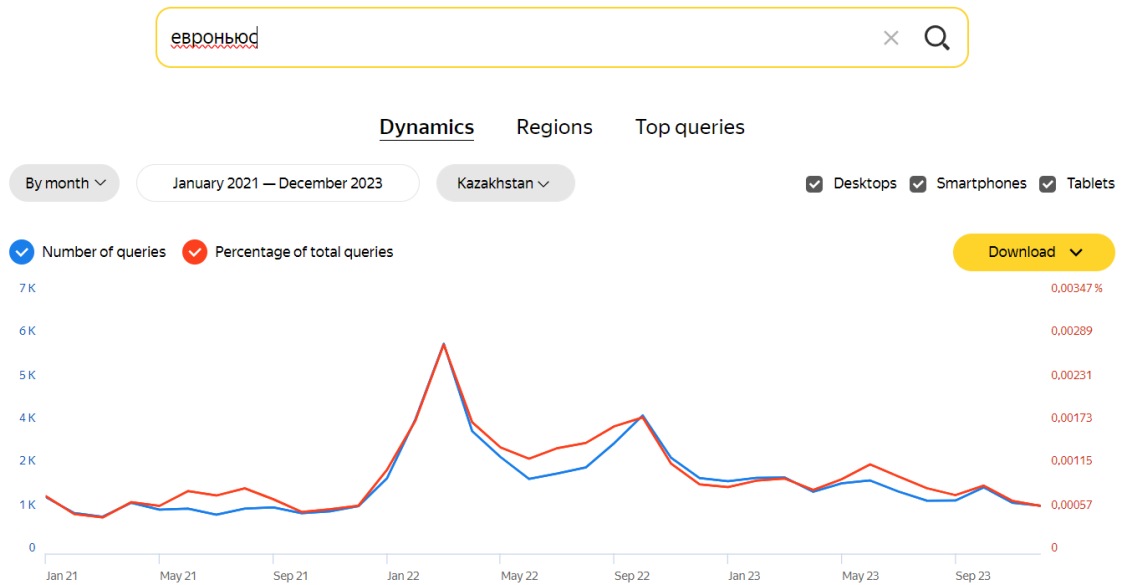
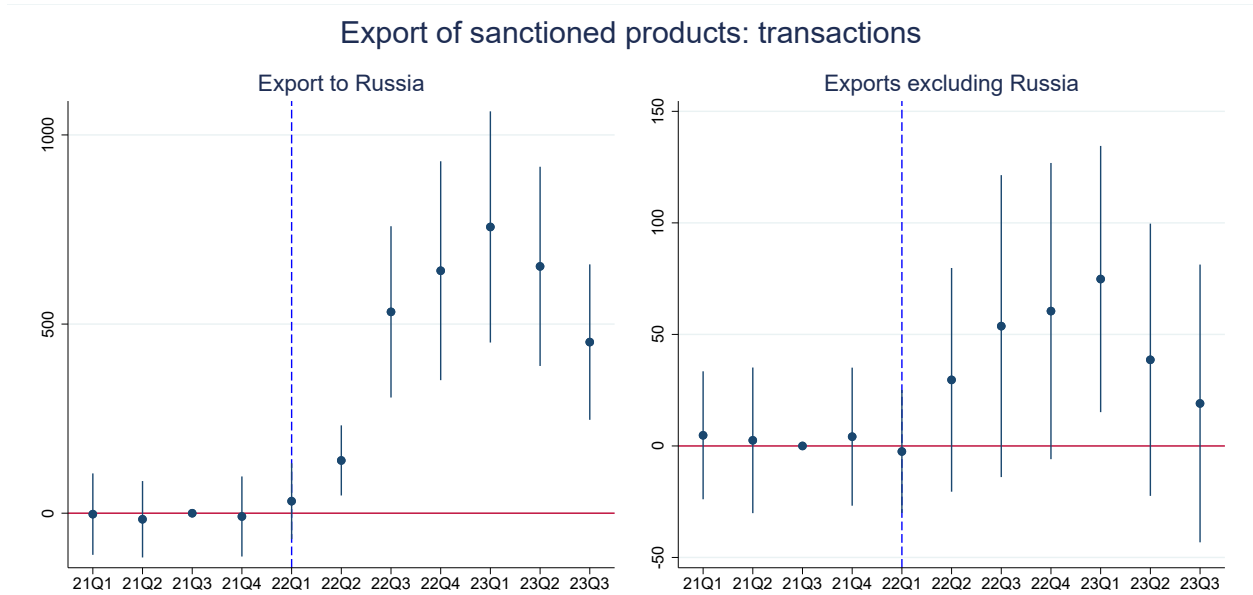


Figure A12: Search for “Euronews” on Yandex



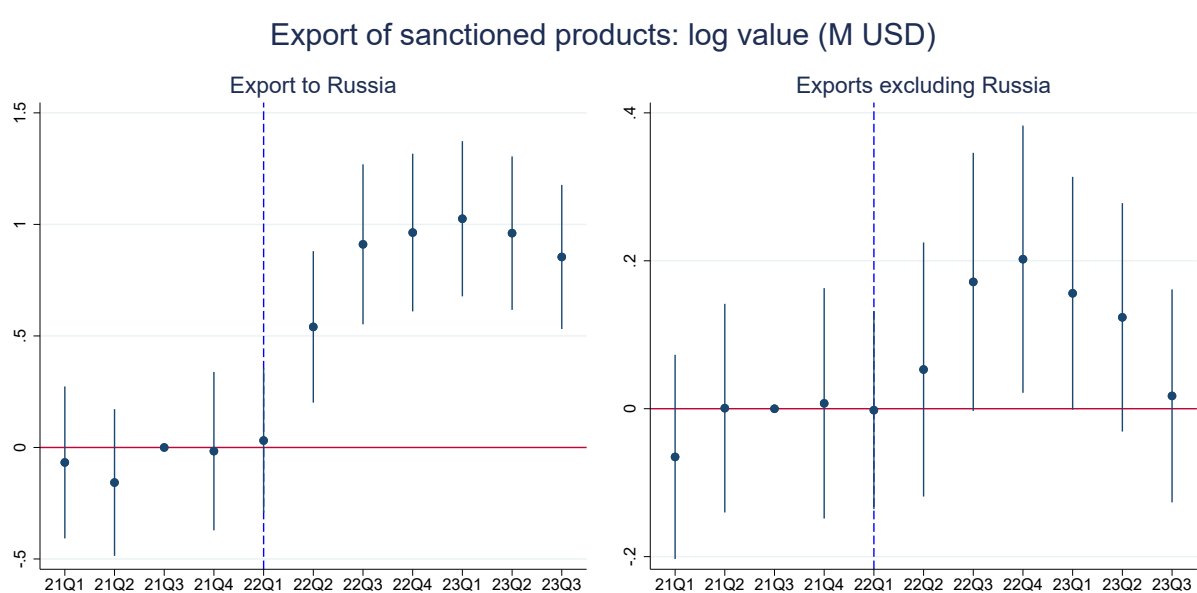
B Appendix

Figure B1: The number of export transactions growth for sanctioned products: event-study



Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B2: The log of the export value growth for sanctioned products: event-study



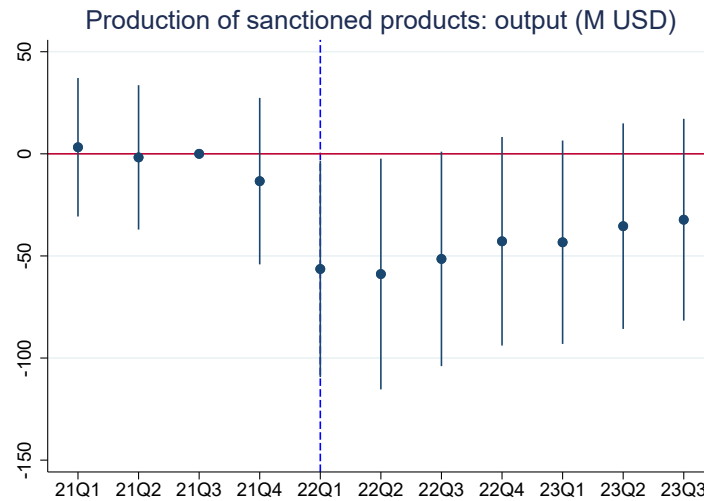
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B3: The log of the number of export transactions growth for sanctioned products: event-study



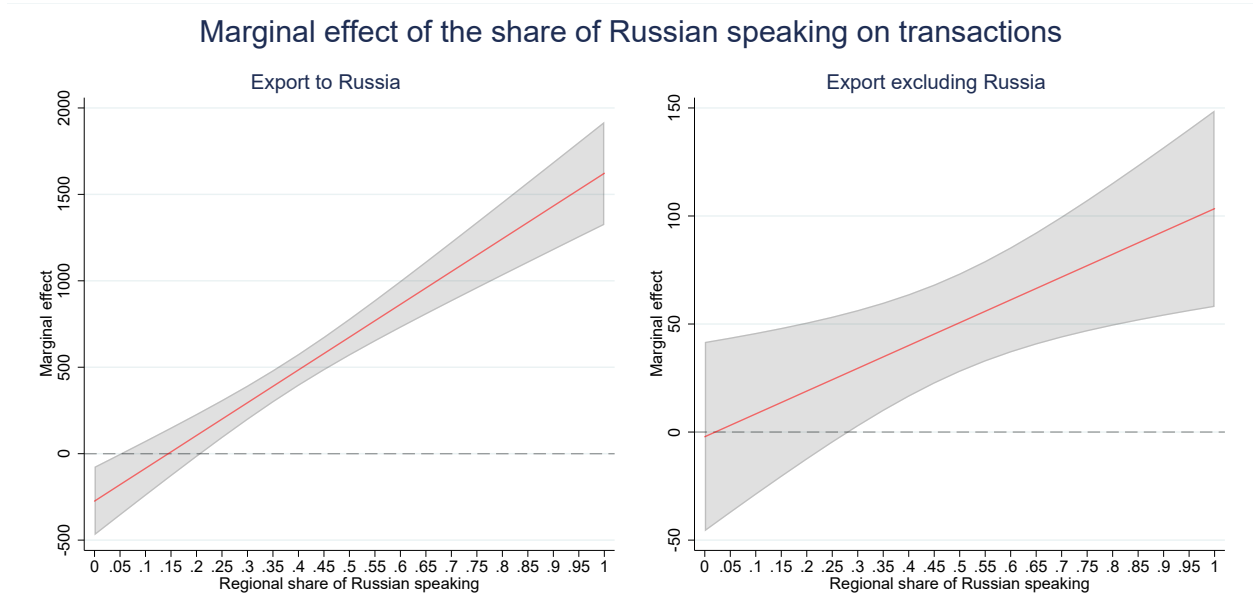
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B4: The output change for sanctioned products: event-study



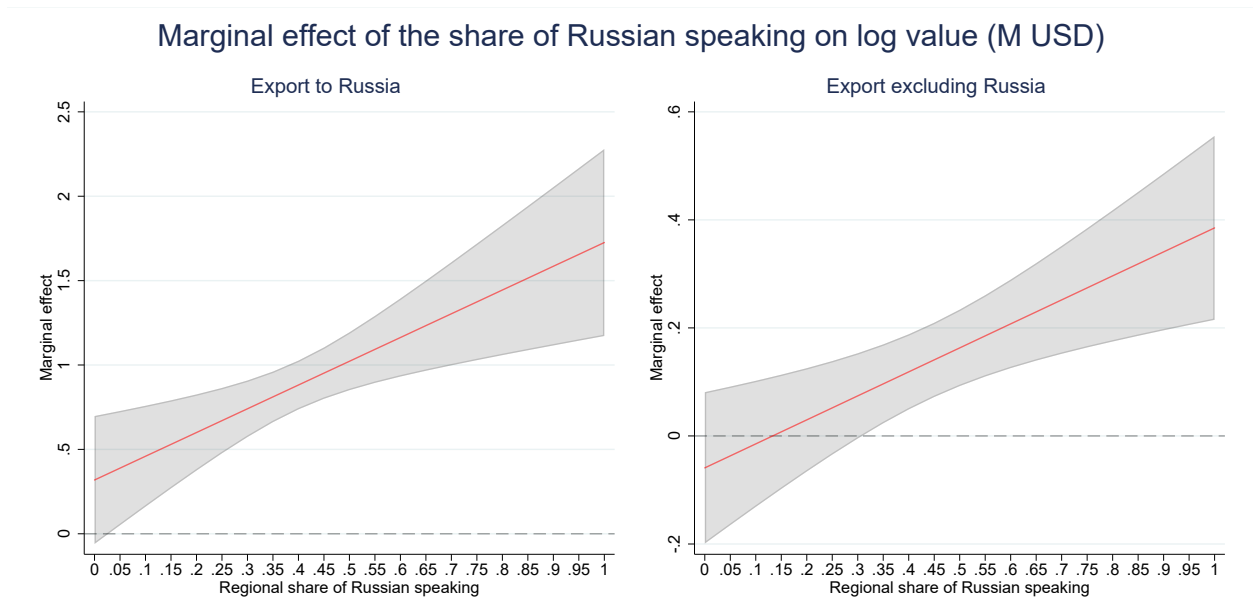
Note. The graph shows the event-study coefficients from (3). Blue dots show the coefficients β_τ . The 95% confidence intervals are calculated w.r.t. robust standard errors.

Figure B5: The number of transactions growth for sanctioned products by regional share of Russian-speaking Kazakh



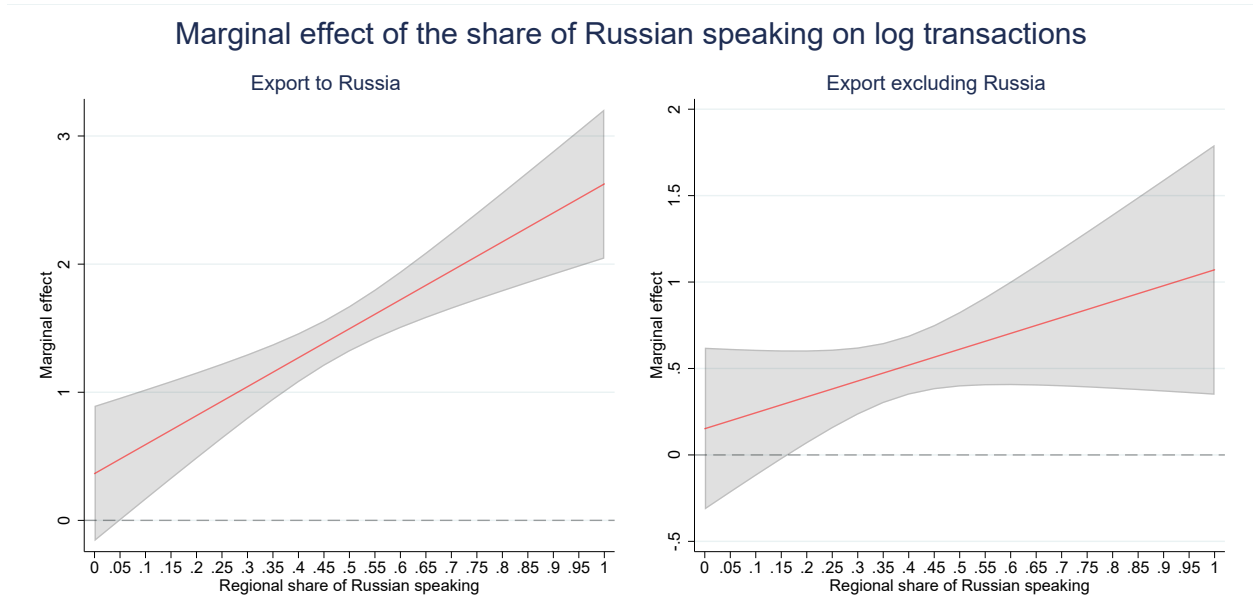
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The dependent variable is the number of transactions for a product class in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B6: The log of the value of export growth for sanctioned products by regional share of Russian-speaking Kazakh



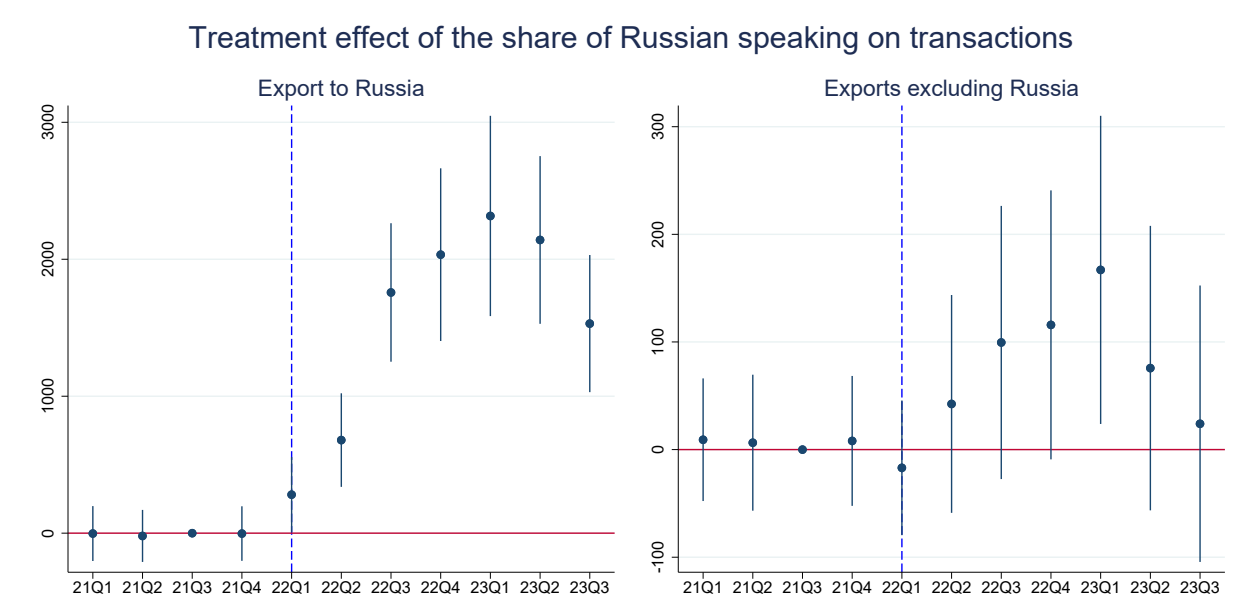
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The dependent variable is the log of export value for a product class in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B7: The log of the number of transactions growth for sanctioned products by regional share of Russian-speaking Kazakh



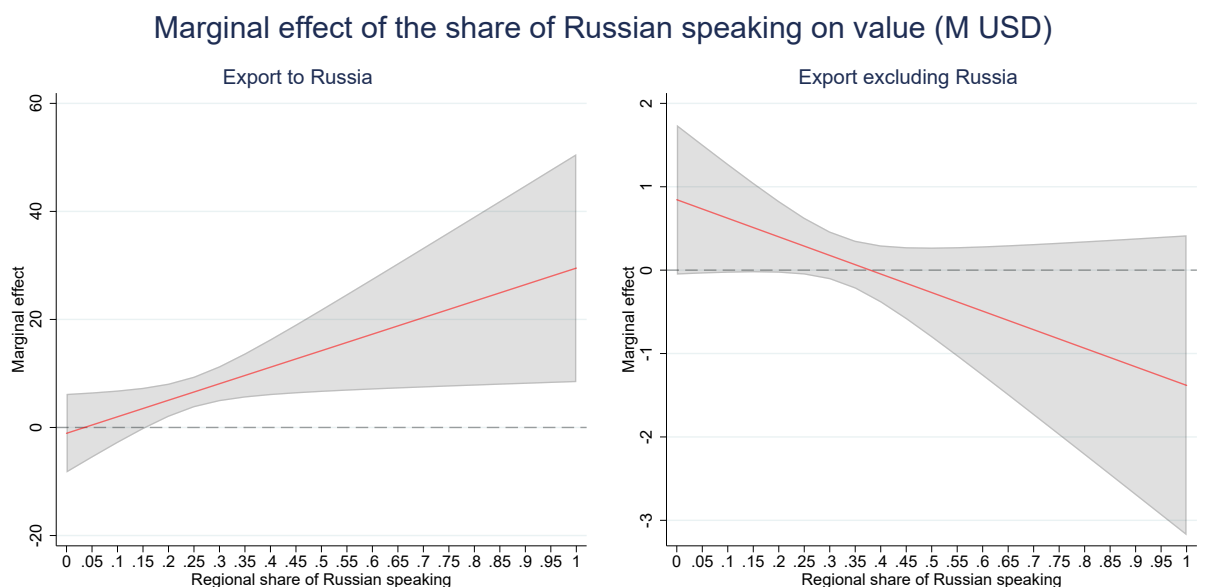
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The dependent variable is the log of the number of transactions for a product class in a region-month. The regional share of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B8: Event study for the transactions growth by regional share of Russian-speaking Kazakh



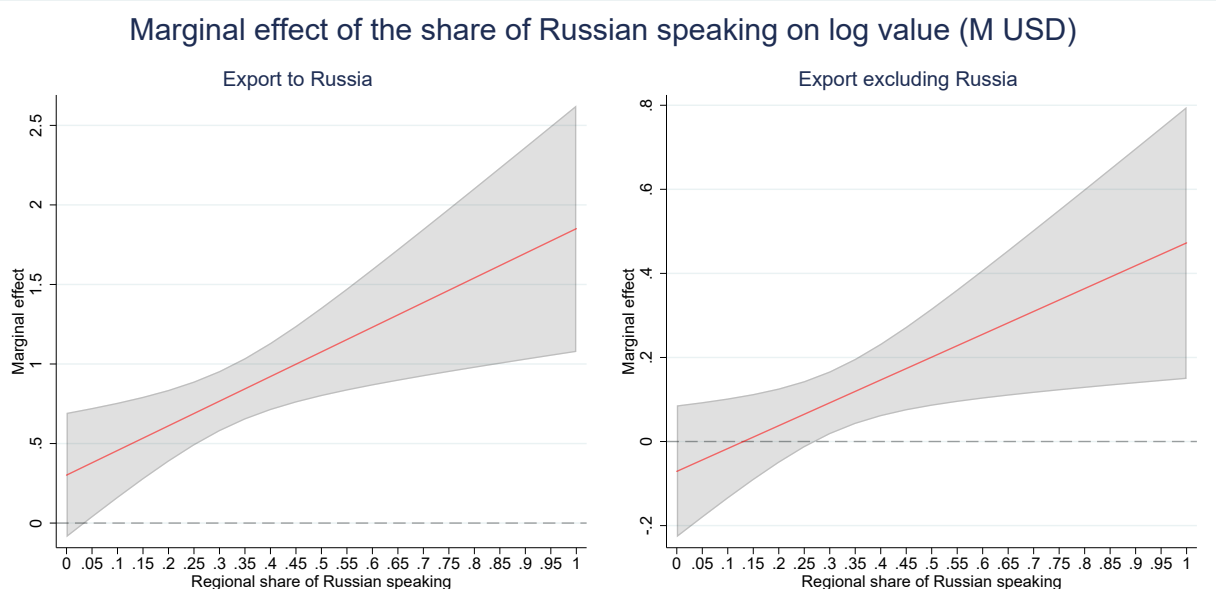
Note. The figure shows coefficients γ_τ from (5). The dependent variable is the number of transactions for a product class in a region-month.

Figure B9: The value of export growth for sanctioned products by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



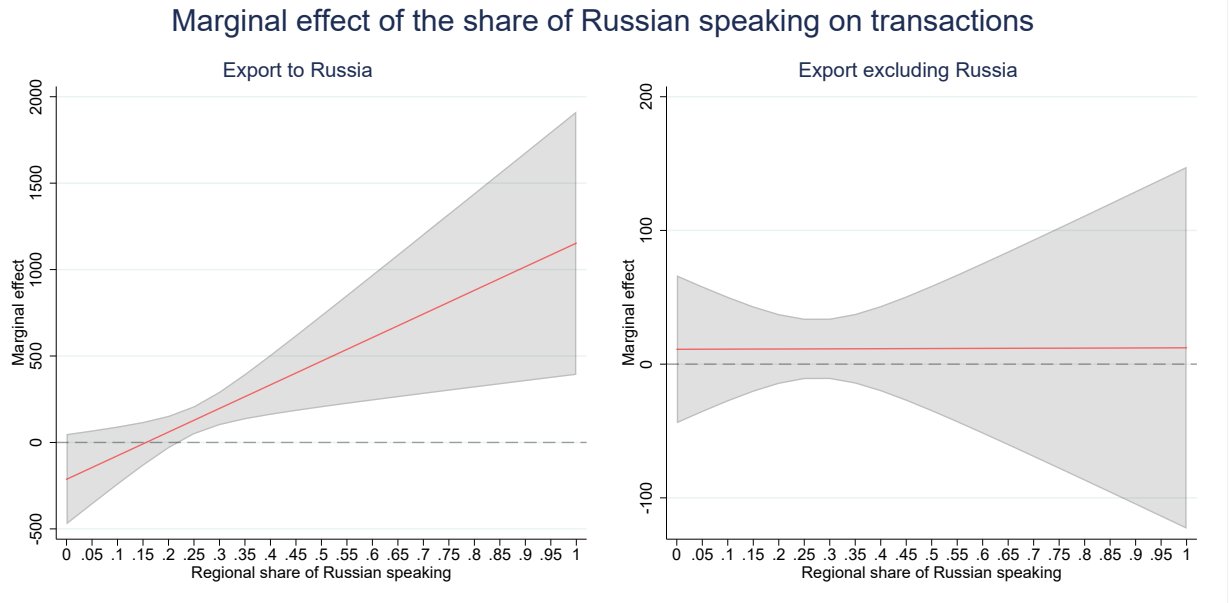
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (6) that includes regional share of ethnic Russians. The dependent variable is the export value for a product class in a region-month. The regional shares of Russian-speaking Kazakhs and ethnic Russians are based on the 2021 Census.

Figure B10: The log of the value of export growth for sanctioned products by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



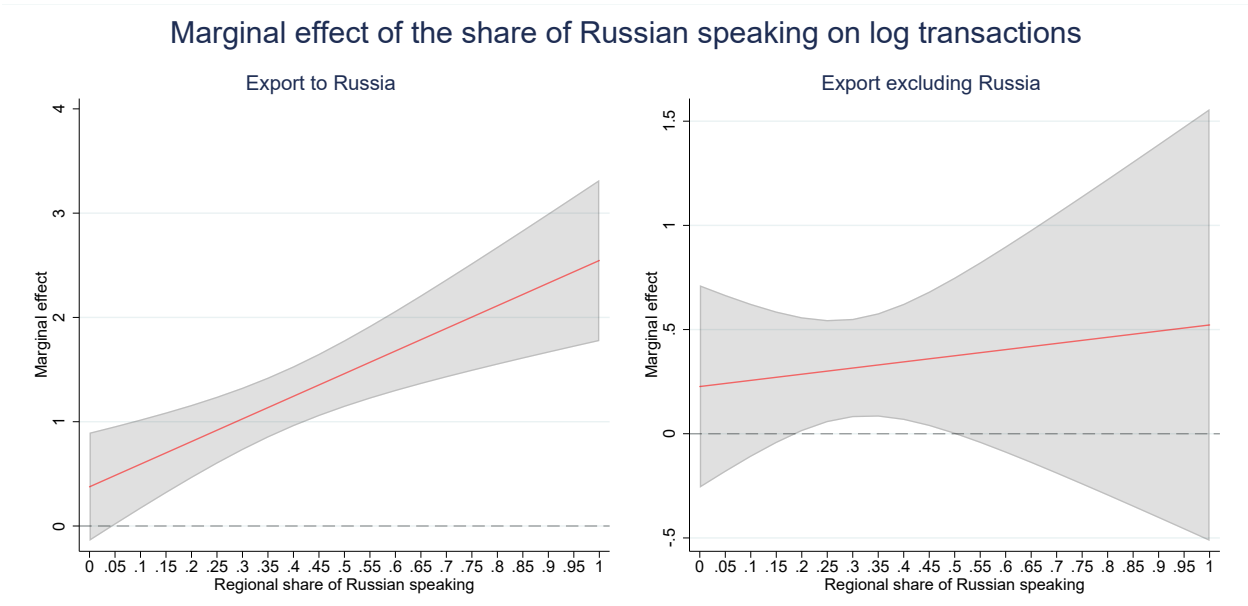
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (6) that includes regional share of ethnic Russians. The dependent variable is the log of the export value for a product class in a region-month. The regional shares of Russian-speaking Kazakhs and ethnic Russians are based on the 2021 Census.

Figure B11: Transactions growth for sanctioned products by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



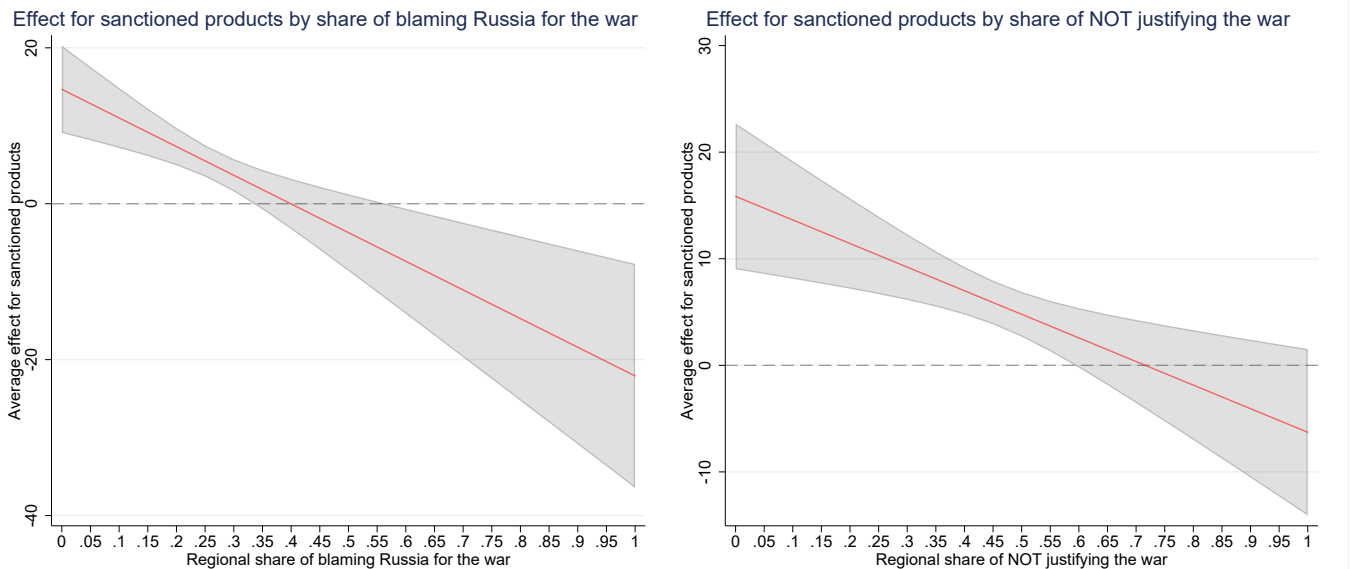
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (6) that includes regional share of ethnic Russians. The dependent variable is the number of transactions for a product class in a region-month. The regional shares of Russian-speaking Kazakhs and ethnic Russians are based on the 2021 Census.

Figure B12: The log of transactions growth for sanctioned products by regional share of Russian-speaking Kazakh, net of regional share of ethnic Russians



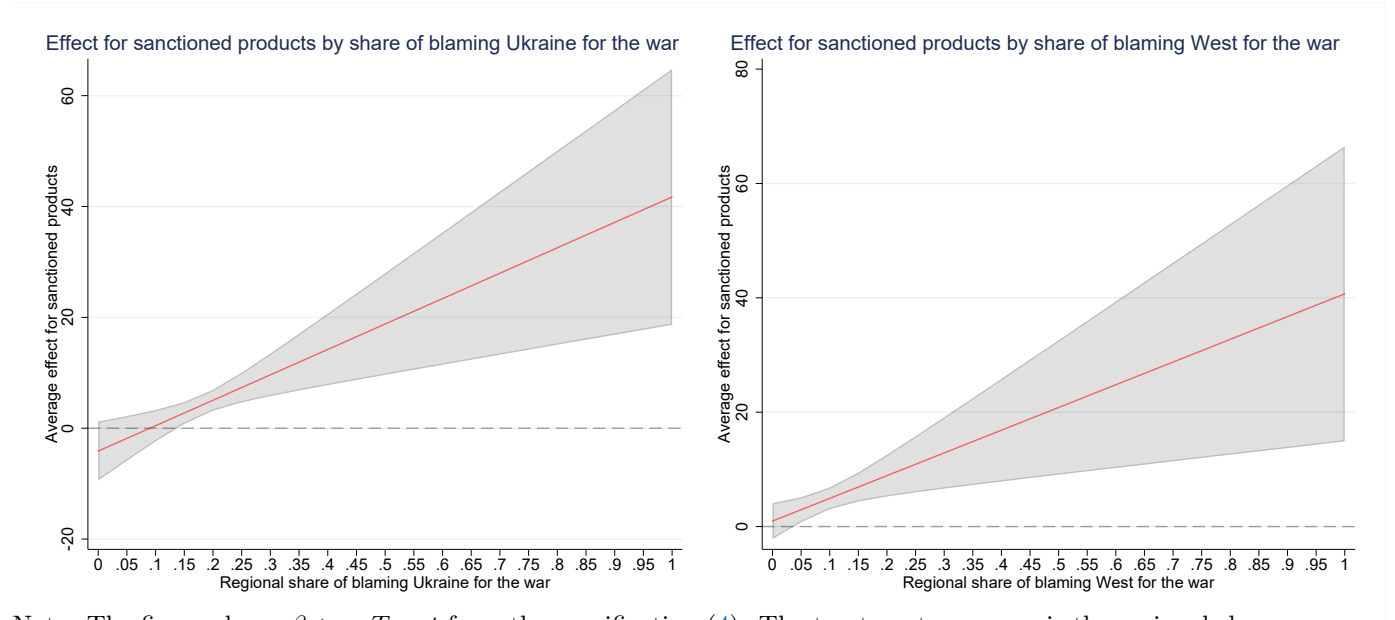
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (6) that includes regional share of ethnic Russians. The dependent variable is the log of the number of transactions for a product class in a region-month. The regional shares of Russian-speaking Kazakhs and ethnic Russians are based on the 2021 Census.

Figure B13: The value of export growth for sanctioned products by regional share blaming Russia for the war and by regional share not justifying the war



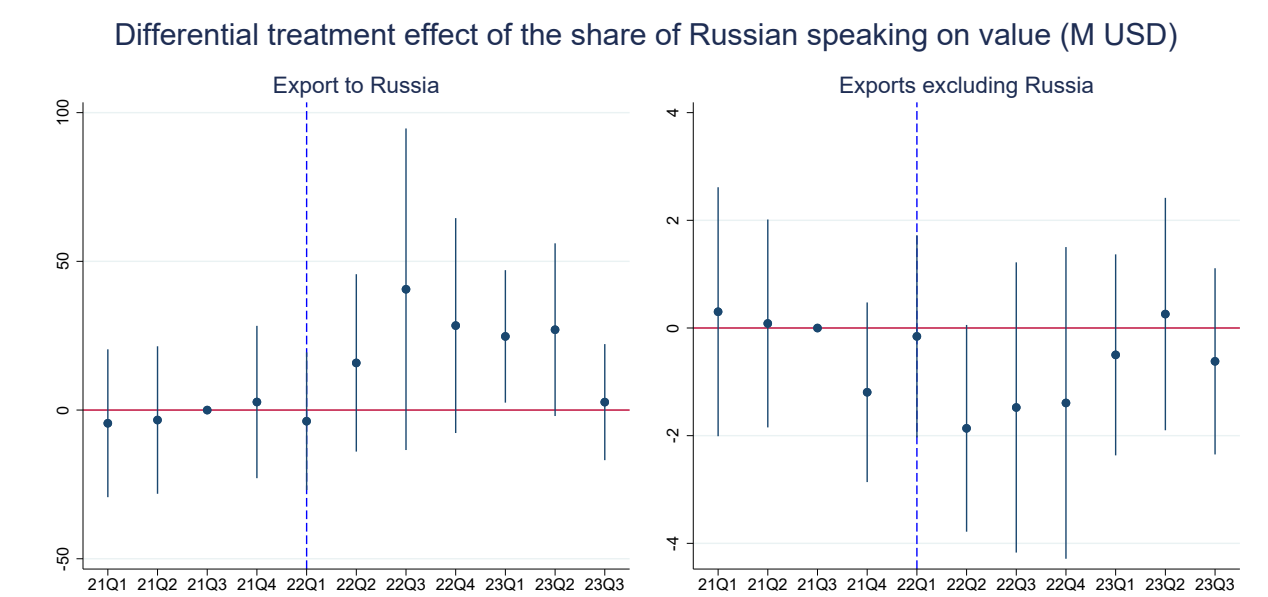
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The treatment exposure is the regional share of respondents blaming Russia for the war (left) and the regional share of respondents not justifying the war (right), both from the CAB surveys. The dependent variable is the export value for a product class in a region-month (M USD).

Figure B14: The value of export growth for sanctioned products by regional share blaming Ukraine and by regional share blaming the West for the war



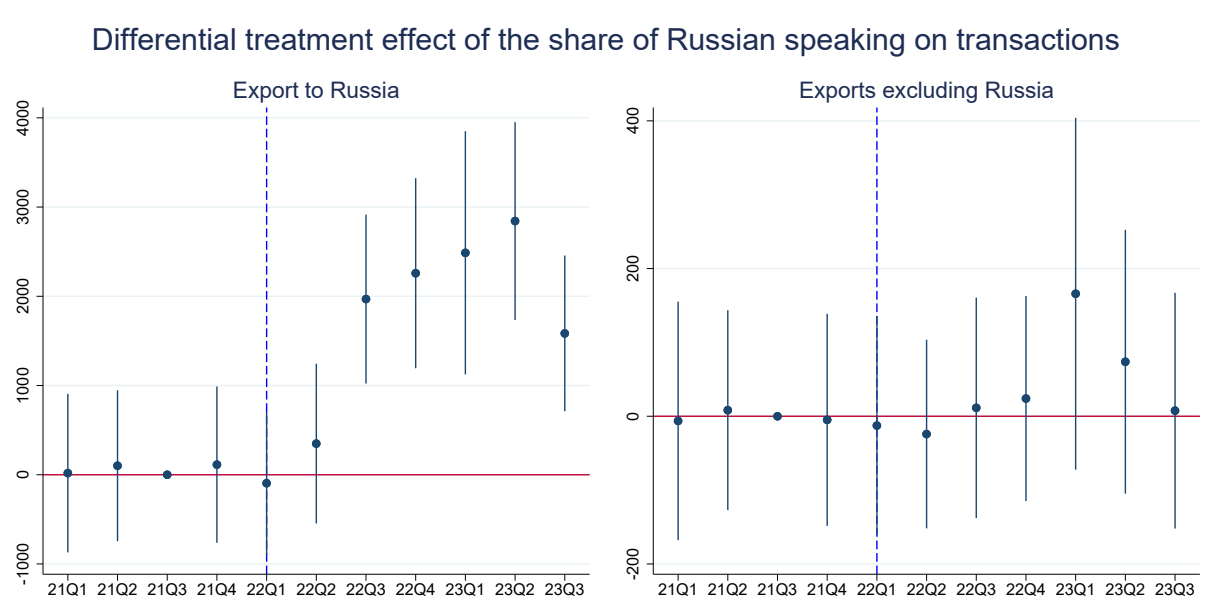
Note. The figure shows $\beta + \gamma \cdot Treat$ from the specification (4). The treatment exposure is the regional share of respondents blaming Ukraine (left) and the regional share of respondents blaming the West (right) for the war, both from the CAB surveys. The dependent variable is the export value for a product class in a region-month (M USD).

Figure B15: The event study of the effect of Russian language on the export value growth



Note. The figure shows the event study results stem from the DID specification (7) (i.e. γ_τ). The dependent variable is the export value for sanctioned products in a region-month. The regional shares of Russian-speaking Kazakhs is based on the 2021 Census.

Figure B16: The event study of the effect of Russian language on transactions growth



Note. The figure shows the event study results stem from the DID specification (7) (i.e. γ_τ). The dependent variable is the number of transactions for sanctioned products in a region-month. The regional shares of Russian-speaking Kazakhs is based on the 2021 Census.

C Appendix

Table C1: Summary statistics for the online survey

	(1)		
	count	mean	sd
Ethnic Russian	5025	0.24	0.43
Female	5025	0.51	0.50
Age	5025	42.65	16.13
Number of children	5025	1.26	1.46
Higher education	5025	0.31	0.46
Work full-time or part-time	5025	0.61	0.49
Household welfare (scale 1-5)	5025	2.85	1.14
Internet usage (scale 1-6)	5025	4.47	1.80
Rural area	5025	0.16	0.36
Intro page in Russian	5025	0.49	0.50
Survey in Russian	5025	0.53	0.50
Russian at home	5025	0.38	0.48
Russian with friends	5025	0.41	0.49
News in Russian	5025	0.45	0.50
Sanctions evasion: justified	5025	0.30	0.46
Invasion/SMO: justified	5025	0.26	0.44
Opinion of Russia (scale 1-5)	4334	3.36	1.02
Opinion of China (scale 1-5)	4327	3.28	0.99
Opinion of the USA (scale 1-5)	4366	3.34	1.09
Government trust (scale 1-4)	4340	2.99	0.94
Protest participation: justified	5025	0.32	0.46
Work $\geq$ 50 h.: justified	5025	0.31	0.46

Note. The table summarizes the characteristics of the participants in the online survey, including demographics, language use, socioeconomic status, attitudes toward Russian and non-Russian affairs.

Table C2: Balance statistics for the language experiment in an online survey

Variable	(1)	(2)	(3)	(4)
	Intro	random page		
	Kazakh	Russian	Difference	Cohen's d
Ethnic Russian	0.238 (0.426)	0.252 (0.434)	0.014 (0.012)	0.023
Female	0.514 (0.500)	0.511 (0.500)	-0.003 (0.014)	-0.004
Age	42.651 (16.276)	42.659 (15.975)	0.008 (0.455)	0.000
Number of children	1.295 (1.480)	1.222 (1.440)	-0.073* (0.041)	-0.035
Higher education	0.306 (0.461)	0.309 (0.462)	0.002 (0.013)	0.004
Work full-time or part-time	0.606 (0.489)	0.619 (0.486)	0.013 (0.014)	0.019
Household welfare (scale 1-5)	2.859 (1.142)	2.845 (1.129)	-0.014 (0.032)	-0.009
Internet usage (scale 1-6)	4.492 (1.789)	4.438 (1.805)	-0.054 (0.051)	-0.021
Rural area	0.155 (0.362)	0.157 (0.364)	0.002 (0.010)	0.004
Observations	2,542	2,483	5,025	

Note. The table shows the balance statistics of the randomly assigned intro page in the online survey for demographics.

Table C3: Russian intro page and attitudes toward non-Russian affairs

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Opinion of China	Opinion of the USA	Government trust	Protest participation	Work $\geq$ 50 h.					
Intro page in Russian	0.051* (0.030)	0.040 (0.029)	0.0025 (0.033)	0.0044 (0.029)	0.047* (0.028)	0.041 (0.027)	-0.0021 (0.013)	-0.0015 (0.012)	-0.0038 (0.013)	-0.0026 (0.012)
Constant	3.25*** (0.021)	3.71*** (0.11)	3.34*** (0.023)	3.90*** (0.11)	2.97*** (0.020)	3.51*** (0.094)	0.32*** (0.0092)	0.55*** (0.048)	0.32*** (0.0092)	0.39*** (0.046)
Observations	4,327	4,327	4,366	4,366	4,340	4,340	5,025	5,025	5,025	5,025
R-squared	0.001	0.080	0.000	0.219	0.001	0.144	0.000	0.112	0.000	0.107
Demographics	N	Y	N	Y	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y	N	Y	N	Y
Estimat.	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the reduced form for the effect of randomized language of the intro page on attitudes toward non-Russian affairs. The dependent scale variables are the opinion of China (Columns 1-2) and the USA (Columns 3-4), ranging from 1 – “Very negative” to 5 – “Very positive”. The dependent scale variables are the standard generalized trust question (Columns 5-6) and trust in the government (Columns 7-8), ranging from 1 – “Cannot be trusted at all” to 4 – “Can be trusted completely”. In Columns 9-10, the dependent binary variable equals 1 if protest participation for political changes is “justified”. In Columns 11-12, the dependent binary variable equals 1 if working more than 50 hours per week is “justified”. Odd columns do not have controls. Even columns include additional controls: ethnicity, gender, age, number of kids, education, occupation, household welfare, internet use intensity, a rural binary indicator, and regional fixed effects. Robust standard errors are in parentheses.

Table C4: The effect of Russian survey language on attitudes toward non-Russian affairs

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Opinion of China		Opinion of the USA		Government trust		Protest participation		Work $\geq$ 50 h.	
Survey in Russian	0.21*	0.17	0.011	0.019	0.19	0.17	-0.0095	-0.0068	-0.017	-0.012
	(0.13)	(0.13)	(0.14)	(0.13)	(0.12)	(0.11)	(0.058)	(0.058)	(0.058)	(0.058)
Constant	3.17***	3.68***	3.33***	3.89***	2.90***	3.48***	0.32***	0.55***	0.32***	0.39***
	(0.066)	(0.12)	(0.070)	(0.12)	(0.058)	(0.10)	(0.032)	(0.051)	(0.032)	(0.049)
Observations	4,327	4,327	4,366	4,366	4,340	4,340	5,025	5,025	5,025	5,025
Demographics	N	Y	N	Y	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y	N	Y	N	Y
Estimat.	IV	IV	IV	IV	IV	IV	IV	IV	IV	IV
Kleibergen-Paap Wald rk F	261.9	383.9	270.2	387.9	285.2	407.7	269.3	408.7	269.3	408.7

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the IV results of the effect of completing the survey in Russian on attitudes toward non-Russian affairs. The dependent variables (and control variables in the even columns) are the same as in Table C3. The heteroskedasticity-robust F-statistic shows that the instrument is strong (≥ 20) for standard inference (Andrews et al. 2019). Robust standard errors are in parentheses.

Table C5: The effect of Russian survey language on Russian language use

VARIABLES	(1) Russian at home	(2) Russian with friends	(3) Russian with friends	(4) Russian with friends	(5) News in Russian	(6) News in Russian
Survey in Russian	0.055 (0.059)	0.010 (0.042)	0.14** (0.057)	0.11** (0.043)	0.14** (0.057)	0.11** (0.046)
Constant	0.35*** (0.032)	0.15*** (0.038)	0.34*** (0.031)	0.055 (0.039)	0.37*** (0.031)	0.052 (0.042)
Observations	5,025	5,025	5,025	5,025	5,025	5,025
Demographics	N	Y	N	Y	N	Y
Region FE	N	Y	N	Y	N	Y
Estimat.	IV	IV	IV	IV	IV	IV
Kleibergen-Paap Wald rk F	269.3	408.7	269.3	408.7	269.3	408.7

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the IV results for the effect of completing the survey in Russian on Russian-language use across different circumstances. The heteroskedasticity-robust F-statistic shows that the instrument is strong (≥ 20) for standard inference ([Andrews et al. 2019](#)). Robust standard errors are in parentheses.

D Appendix

D.1 Language Habits and Justification of the War in Ukraine

We begin by examining the relationship between linguistic habit in speaking Russian and attitudes toward the war in Ukraine. We use four CAB survey waves conducted after February 2022 and consider the following two-way fixed-effects linear specification:

$$y_{irt} = \alpha L_{irt} + \mathbf{X}_{irt}\beta + \kappa_r + \lambda_t + \epsilon_{irt} \quad (14)$$

The subscript i denotes a respondent from region r and wave t .³⁵ The binary variable $L_{irt} = 1$ if a respondent speaks Russian as their daily language at home, and 0 otherwise. The vector $\mathbf{X}_{irt}$ includes the following set of control variables: gender, five age categories, a dummy variable for education at the university level or higher, five categories of household size, dummy variables for urban/rural settlement and ethnicity (Russian and Kazakh, compared to others), as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey. Note that data on ethnicity enables us to separate language habits from ethnic background. The variables κ_r and λ_t are the region and survey wave fixed effects, respectively. As dependent variables y_{irt} , we consider answers to the following two questions about the war in Ukraine: (i) the extent to which the SMO is justified (both as a binary and a scaled measure) and (ii) who is mainly responsible for the war (Russia, Ukraine or the West). Our primary focus is on the coefficient α , which shows the association between attitudes toward the war and language habits, while controlling for socio-demographics and ethnicity.

Table D1 shows the results for the binary measure “the SMO is justified” for the whole sample (column 1) and for different relevant sub-samples: column 2 zooms in on firm heads, columns 3 and 4 on respondents older or younger than 40 years of age, column 5 looks at respondents who indicated to pay a lot of attention to information about the war, and column 6 considers ethnically non-Russian respondents. We find that Russian-speaking respondents are 11 p.p. more likely to agree that SMO is justified, which is equivalent to 42% when referring to the sample average. For all the above-mentioned sub-samples, the results are similar, with

³⁵Regions are three major cities, Astana, Almaty, and Shymkent, and 17 other sub-national regions.

some heterogeneity by age. Notably, even among the group of ethnic non-Russians, those who mainly speak Russian in their daily lives are 10 p.p. more likely to justify the war than those who mainly speak Kazakh. Table D6 in Appendix B provides the regression output demonstrating all control variables. Table D7 in Appendix B shows the estimates for the non-binary outcome variable. The results are essentially the same, except for the result for firm heads being no longer significant. Finally, Table D8 in Appendix B uses “the SMO is not justified” as a binary dependent variable, showing that Russian speakers are about 8 p.p. less likely to condemn the war than non-Russian speakers.

Table D1: Russian language habit and war justification

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	SMO is justified (binary)					
Russian language	0.11*** (0.015)	0.090** (0.037)	0.16*** (0.026)	0.077*** (0.019)	0.13*** (0.019)	0.10*** (0.015)
Observations	6,020	955	2,366	3,654	3,889	4,864
R-squared	0.140	0.154	0.223	0.082	0.178	0.078
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the regression results (14) for a binary dependent variable, “SMO is justified”. Robust standard errors are in parentheses.

As the next step, we consider a set of binary variables that assign responsibility for the war to three possible actors — Russia, Ukraine, or the West — as the dependent variable y_{irt} in equation (14). Table D2 shows the results with a breakdown by dependent variables and sub-groups of respondents. Column 1 of panel A shows that, in general, Russian-speaking respondents are 5.7 p.p. less likely to blame Russia for being the main culprit of the war, which is equivalent to 21.7% when referring to the sample average. The association is even stronger for respondents who are particularly interested in information about the war (column 5 of panel A), while for firm heads and younger respondents, it is not significant (columns 2 and 4 of panel A). Panel B shows that Russian-speakers are not more likely than non-Russian speakers to assign responsibility for the war to Ukraine. However, Panel C shows that Russian speakers are 8.7 p.p. more likely to blame the West, which is equivalent to 72% when referring

to the sample average. This association is significant for all sub-samples, including firm heads (column 2 of panel C) and younger respondents (column 4 of panel C).

Table D2: Russian language habit and responsibility for the situation in Ukraine

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. War responsibility: Russia						
Russian language	-0.057*** (0.017)	-0.025 (0.042)	-0.094*** (0.028)	-0.034 (0.021)	-0.070*** (0.022)	-0.058*** (0.017)
Panel B. War responsibility: Ukraine						
Russian language	0.013 (0.014)	-0.028 (0.033)	0.028 (0.024)	0.0058 (0.018)	0.0012 (0.018)	0.013 (0.014)
Panel C. War responsibility: US, EU, NATO						
Russian language	0.087*** (0.012)	0.079*** (0.031)	0.13*** (0.024)	0.063*** (0.013)	0.10*** (0.016)	0.083*** (0.012)
Observations	6,020	955	2,366	3,654	3,889	4,864
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the results of regression (14) for binary dependent variables assigning responsibility for the war to Russia, Ukraine or the West. Robust standard errors are in parentheses.

We conclude that Russian-speaking respondents are more likely to justify the war, are less likely to blame Russia, and are more likely to blame the West as compared to non-Russian speakers.³⁶ These results underline the critical role of language habits in shaping political views, in addition to ethnicity and other factors.

D.2 Language Habits and Media Consumption

In this section, we examine a set of *potential mechanisms* that explain why Russian speakers are more likely to justify the war, a finding that holds even for the part of the Kazakh popula-

³⁶Figure D1 in Appendix B shows the dynamics of the association between language habit and war attitudes (i.e. coefficients α_t for $L_{irt} \times \lambda_t$ keeping other controls from specification (14).) All war attitude variables exhibit a strong association with the language in the first wave and stabilization at a lower magnitude afterward.

tion that is ethnically non-Russian. We consider information channels that have traditionally played a significant role in Central Asia, namely, Russian traditional media (TV and Radio) and labor migration. We use them as dependent variables in equation (14).

Table D3: Russian language habit and channels for Russian propaganda

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A. News from Russian TV, radio, newspapers					
Russian language	0.022*** (0.0063)	0.0074 (0.012)	0.037*** (0.014)	0.010* (0.0060)	0.029*** (0.0089)	0.016*** (0.0061)
	Panel B. Absence of remittances					
Russian language	0.0082 (0.010)	0.024 (0.025)	0.0033 (0.016)	0.010 (0.014)	0.0032 (0.012)	0.013 (0.011)
	Panel C. News from the Internet					
Russian language	0.038** (0.016)	-0.017 (0.036)	0.0063 (0.030)	0.052*** (0.020)	0.046** (0.019)	0.038** (0.017)
Observations	6,020	955	2,366	3,654	3,889	4,864
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the results of regression (14) for binary dependent variables indicating consumption of international news from Russian TV, Radio, Newspapers and from the Internet, as well as binary dependent variables indicating that respondents received remittances. Robust standard errors are in parentheses.

Panel A of Table D3 shows that Russian-speaking respondents are more likely than non-Russian speakers to consume Russian traditional media (TV news, radio, and newspapers). This association is stronger for elder respondents and those who pay special attention to information about the war. However, a relatively small share of the population uses Russian traditional media as the main source of information (see Table D4).

There is no evidence that Russian speakers are more or less likely to receive remittances from Russia (Panel B of Table D3), suggesting that this is unlikely to be a channel for the difference in political attitudes.

Table D4: Descriptive statistics for media consumption

	2021	2022	2023
Panel A: Main news source for events outside Kazakhstan			
Internet	59.4	64.6	67
National/local TV, radio, newspapers	10.6	12.2	10.1
Russian TV, radio, newspapers	7.2	5	4.4
Panel B: Russian media consumption			
Use Russian media		65.5	62.1
Trust in news from Russia			36.8
Panel C: Messenger usage (most often)			
Use Telegram	6	8.5	10.1
Use Whatsapp	88.3	87.5	86.7
Panel D: Social media usage (most often)			
Use VK or OK	16.2	11.6	9.2
Use Facebook	8.7	8.2	7.7
Use Instagram	50.8	48.8	47.4
Use Tiktok	7.7	17.1	22.5
Panel E: Reliance on remittance from abroad			
Absence of remittance	88.3	91.9	92.9

Note. Table shows the percentage of media usage from six CAB survey waves for 2021–2023. No weights are applied. The row “*Trust in news from Russia*” shows the share of respondents from September 2023 who strongly or somewhat trust news from Russia. The row “*Absence of remittance*” shows the share of the households that do not rely on remittance. All the percentages are calculated as shares of respondents in the corresponding years without re-weighting.

Panel C of Table D3 then examines Russian-speaking respondents who primarily consume news from the Internet (the main source of information for citizens of Kazakhstan, as shown in Table D4). We see that Russian-speaking respondents are 3.8 p.p. more likely to consume internet news than non-Russian speakers, with the effect being especially prominent for younger respondents. The combined results from panels A and C suggest that while traditional Russian media still play a role for older respondents, younger respondents with similar language habits get their information mostly online.³⁷

The CAB survey only includes a question about news consumption from the Internet,

³⁷Appendix B shows that Russian speakers are more likely to use the internet in daily life (Tables D9 and D10), less likely to consume Kazakh national media for international news (Table D11) and more likely to consume Russian media for news and entertainment (Table D12). Ethnic non-Russians speaking Russian are more likely to trust Russian news (Table D13).

without asking for news consumption from social media or specific websites. However, there is a question about the preferred social media that respondents use most frequently. Combining this question with the question about internet news consumption enables us to study the role of specific social media platforms as channels for information. We consider eight different social media platforms. Vkontakte (VK) and Odnoklassniki (OK) are platforms controlled by the Russian government, with VK being more popular among younger people and OK among older people. Since February 2022, VK, OK, and Telegram have been intensely used by Russian government propaganda. Telegram, in particular, has become a platform where pro-Kremlin bloggers with large audiences are providing information about the war from a pro-Kremlin perspective.³⁸ Facebook and Instagram, on the other hand, are blocked in Russia. TikTok hosts entertainment content, while WhatsApp is primarily used as a messaging app; therefore, these platforms are less relevant as channels for Russian propaganda.

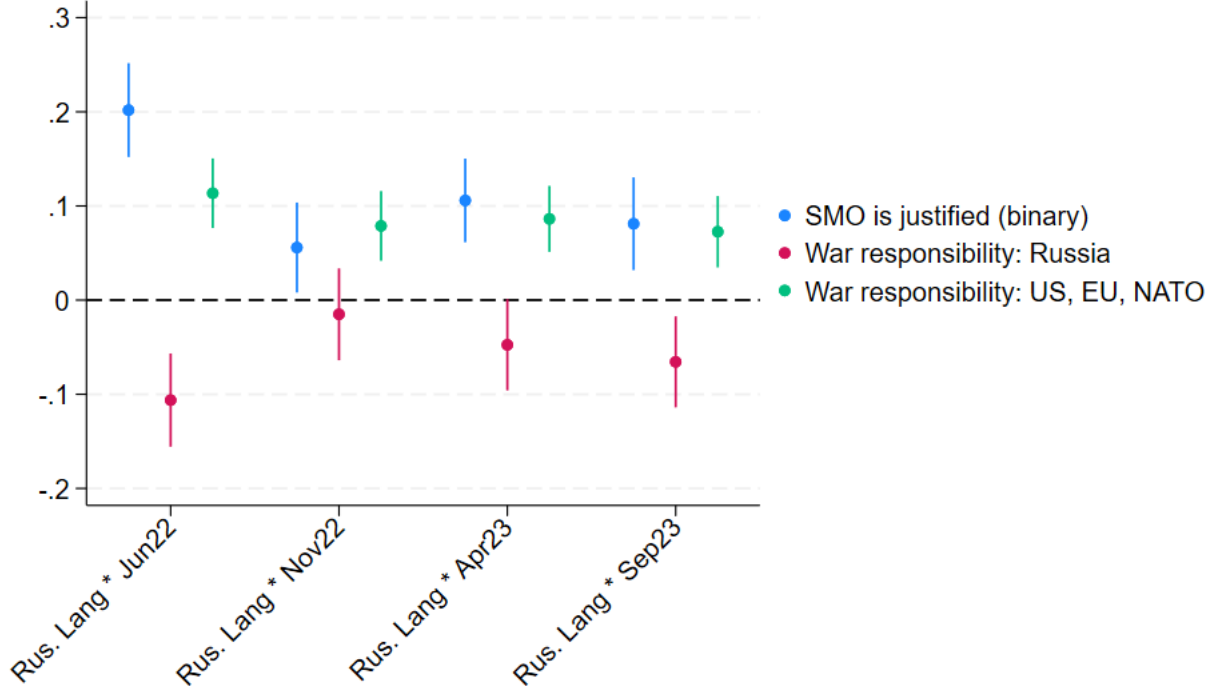
Columns 1 and 2 of Table D5 show that Russian-speaking respondents are significantly more likely to use VK, OK, or Telegram together with the Internet news, while they are less likely to use platforms that the Russian government does not control (columns 3 to 6). These results suggest that online news consumption from Russian social media platforms could be one potential channel driving the difference in war attitudes we identify above.

³⁸See e.g. [Russian war bloggers: Pawns in a political game? DW, 2023](#).

Table D5: Russian language habit, social media and internet news

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: Internet news and the following Social Media:			Social Media:		
VARIABLES	VK or OK	Telegram	Instagram	TikTok	Facebook	WhatsApp
Russian language	0.055*** (0.0063)	0.028*** (0.0068)	-0.026** (0.012)	-0.027*** (0.0085)	-0.040*** (0.0058)	-0.045*** (0.013)
Observations	6,020	6,020	6,020	6,020	6,020	6,020
R-squared	0.014	0.003	0.001	0.002	0.007	0.002
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	All	All	All	All	All
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the results of regression (14) for a set of binary dependent variables that are a combination of using the Internet for international news and frequently using one of the indicated social media platforms. Robust standard errors are in parentheses.

Figure D1: The dynamics of association between the Russian language habit and war attitudes

Note. The figure shows the dynamics of the association between Russian language habit and war attitudes, i.e. coefficients α_t for $L_{irt} \times \lambda_t$ keeping other controls from specification (14). The dependent are binary variables “SMO justification”, “Russia is mainly responsible for the war”, “West is mainly responsible for the war”.

Table D6: Russian language habit and war justification

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	SMO is justified (bianary)					
Russian language	0.11*** (0.015)	0.090** (0.037)	0.16*** (0.026)	0.077*** (0.019)	0.13*** (0.019)	0.10*** (0.015)
Female	-0.018* (0.011)	-0.012 (0.031)	-0.056*** (0.017)	0.0056 (0.014)	-0.012 (0.014)	-0.012 (0.011)
30-39 years old	0.0072 (0.014)	-0.052 (0.034)		0.016 (0.014)	0.021 (0.018)	0.0073 (0.014)
40-49 years old	0.045*** (0.015)	0.043 (0.039)			0.054*** (0.020)	0.020 (0.016)
50-59 years old	0.043** (0.018)	0.015 (0.053)	-0.00090 (0.020)		0.046** (0.022)	0.012 (0.019)
60+ years old	0.13*** (0.021)	0.013 (0.088)	0.084*** (0.022)		0.13*** (0.024)	0.053** (0.023)
Higher education	-0.040*** (0.011)	-0.074** (0.029)	-0.030* (0.018)	-0.049*** (0.014)	-0.054*** (0.014)	-0.041*** (0.011)
Medium household size	0.0015 (0.012)	0.0095 (0.030)	-0.0066 (0.019)	0.010 (0.015)	0.0021 (0.015)	0.0056 (0.013)
Above medium household size	0.016 (0.019)	0.015 (0.050)	-0.024 (0.032)	0.037 (0.024)	0.030 (0.024)	0.011 (0.019)
Large household size	-0.018 (0.033)	0.053 (0.092)	-0.056 (0.052)	0.00076 (0.042)	-0.022 (0.043)	-0.0088 (0.033)
Missing household size	0.0051 (0.037)	0.12 (0.098)	-0.042 (0.067)	0.026 (0.045)	0.071 (0.058)	-0.014 (0.037)
Rural area	0.015 (0.013)	-0.016 (0.041)	-0.024 (0.021)	0.047*** (0.018)	-0.0036 (0.017)	0.015 (0.014)
Ethnically Kazakh	-0.17*** (0.019)	-0.19*** (0.048)	-0.20*** (0.031)	-0.14*** (0.024)	-0.22*** (0.025)	-0.17*** (0.019)
Ethnically Russian	0.080*** (0.022)	0.042 (0.057)	0.11*** (0.034)	0.034 (0.029)	0.065** (0.029)	
Understood most of the questions	0.012 (0.015)	0.020 (0.038)	0.019 (0.023)	0.0081 (0.019)	0.026 (0.018)	0.023 (0.015)
Understood most of the questions but with some help	0.015 (0.023)	0.12 (0.078)	0.0074 (0.035)	0.020 (0.030)	0.017 (0.030)	0.022 (0.023)
Had difficulty understanding most of the questions, even with help	-0.042 (0.043)	-0.073 (0.12)	0.023 (0.075)	-0.10** (0.040)	-0.047 (0.064)	-0.045 (0.041)
Was comfortable with most of the questions	-0.035** (0.014)	-0.0052 (0.036)	-0.050** (0.022)	-0.027 (0.017)	-0.033* (0.017)	-0.037** (0.014)
Was comfortable with only some of the questions	-0.15*** (0.030)	-0.13 (0.087)	-0.15*** (0.051)	-0.14*** (0.036)	-0.14*** (0.047)	-0.12*** (0.031)
Was generally uncomfortable with the survey questionnaire	-0.17*** (0.038)	-0.24*** (0.089)	-0.13** (0.058)	-0.18*** (0.046)	-0.14** (0.062)	-0.13*** (0.039)
Wave 12	-0.049*** (0.015)	-0.080** (0.040)	-0.037 (0.025)	-0.052*** (0.020)	-0.041** (0.019)	-0.020 (0.016)
Wave 13	-0.069*** (0.015)	-0.11*** (0.040)	-0.053** (0.024)	-0.071*** (0.019)	-0.063*** (0.019)	-0.063*** (0.016)
Wave 14	-0.037** (0.016)	-0.031 (0.041)	-0.040 (0.025)	-0.028 (0.021)	-0.027 (0.020)	-0.0089 (0.017)
Constant	0.33*** (0.029)	0.40*** (0.073)	0.39*** (0.047)	0.31*** (0.035)	0.37*** (0.037)	0.33*** (0.029)
Observations	6,020	955	2,366	3,654	3,889	4,864
R-squared	0.140	0.154	0.223	0.082	0.178	0.078
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the regression results (14) for a binary dependent variable, “SMO is justified”. Robust standard errors are in parentheses.

Table D7: Russian language habit and war justification (scale from 1 to 4)

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	SMO is justified (scale)					
Russian language	0.27*** (0.042)	0.13 (0.11)	0.44*** (0.074)	0.16*** (0.051)	0.31*** (0.051)	0.26*** (0.043)
Observations	4,458	740	1,818	2,640	3,200	3,566
R-squared	0.207	0.227	0.309	0.137	0.241	0.133
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the regression results (14) for a discrete dependent variable measured from 1 “SMO is completely unjustified” to 4 “SMO is completely justified”. Responses “No answer” and “Don’t know” are excluded. Robust standard errors are in parentheses.

Table D8: Russian language habit and war condemnation

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	SMO is not justified (binary)					
Russian language	-0.078***	-0.070	-0.11***	-0.053**	-0.10***	-0.077***
Observations	6,020	955	2,366	3,654	3,889	4,864
R-squared	0.123	0.167	0.189	0.096	0.172	0.104
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The table shows the regression results (14) for a binary dependent variable, “SMO is not justified”. Robust standard errors are in parentheses.

Table D9: Russian language habit and daily use of the Internet via a phone

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Daily Internet use					
Russian language	0.042*** (0.012)	0.017 (0.024)	0.040* (0.023)	0.043*** (0.013)	0.047*** (0.014)	0.040*** (0.012)
Observations	6,020	955	2,366	3,654	3,889	4,864
R-squared	0.076	0.089	0.090	0.039	0.094	0.083
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Table shows the results of regression (14) for a binary dependent variable, “Daily use of the Internet via a phone”. Robust standard errors in parentheses.

Table D10: Russian language habit and frequency of Internet use via a phone (scale 1-5)

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Frequency of Internet use (from 1 daily to 5 never)					
Russian language	-0.069*** (0.026)	-0.038 (0.043)	-0.082 (0.054)	-0.061** (0.027)	-0.080*** (0.029)	-0.065** (0.026)
Observations	6,020	955	2,366	3,654	3,889	4,864
R-squared	0.080	0.090	0.089	0.038	0.093	0.089
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians

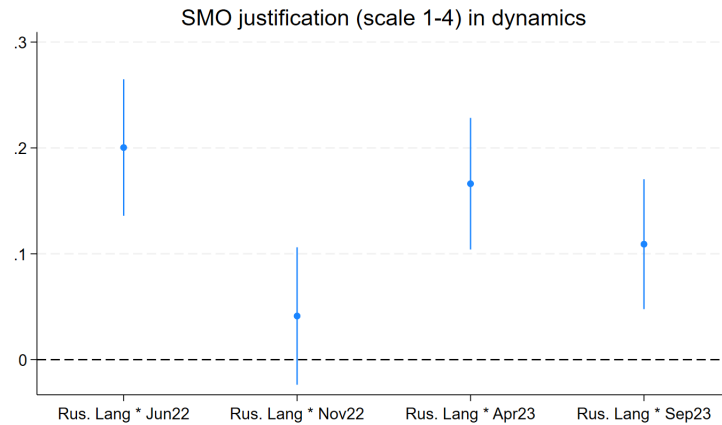
Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Table shows the results of regression (14) for a discrete dependent variable, “Frequency of using the Internet via a phone”, with values from 1 (daily) to 5 (never). Robust standard errors in parentheses.

Table D11: Russian language habit and channels for Russian propaganda: national or local TV, radio, newspapers

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	News from national or local TV, radio, newspapers					
Russian language	-0.036*** (0.010)	-0.012 (0.016)	-0.035 (0.023)	-0.035*** (0.0097)	-0.046*** (0.013)	-0.033*** (0.010)
Observations	6,020	955	2,366	3,654	3,889	4,864
R-squared	0.072	0.061	0.063	0.038	0.089	0.087
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Table shows the results of regression (14) for a binary dependent variable, “News from national or local TV, radio, newspapers”. Robust standard errors in parentheses.

Figure D2: The dynamics of association between the Russian language habit and the ordinal SMO justification measure



Note. Note. The figure shows the dynamics of the association between Russian language habit and war attitudes, i.e. coefficients α_t for $L_{irt} \times \lambda_t$ divided by the mean of dependent variable and keeping other controls from specification (14). The dependent variable is “SMO justification” (scale 1-4). The interpretation of dots is increase in war justification for Russian speakers as the percentage of the mean.

Table D12: Russian language habit and media consumption from different countries for news or entertainment

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Use media from Russia					
Russian language	0.11*** (0.020)	0.039 (0.049)	0.11*** (0.033)	0.11*** (0.026)	0.13*** (0.024)	0.11*** (0.021)
	Use media from the US					
Russian language	0.054*** (0.021)	0.080 (0.050)	-0.036 (0.035)	0.098*** (0.026)	0.047* (0.026)	0.060*** (0.021)
	Use media from Turkey					
Russian language	-0.11*** (0.021)	-0.14*** (0.049)	-0.13*** (0.036)	-0.099*** (0.026)	-0.15*** (0.027)	-0.11*** (0.021)
	Use media from China					
Russian language	-0.018 (0.017)	-0.060 (0.043)	-0.020 (0.031)	-0.018 (0.022)	-0.031 (0.023)	-0.015 (0.018)
Observations	4,511	741	1,768	2,743	2,872	3,611
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Table shows the results of regression (14) for binary dependent variables, “Use media for news or entertainment from Russia/ the US/ Turkey/ China”. Robust standard errors in parentheses.

Table D13: Russian language habit and trust in news or information from different countries

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Trust news or information from Russia					
Russian language	0.070* (0.036)	-0.10 (0.087)	0.10 (0.063)	0.052 (0.045)	0.066 (0.048)	0.076** (0.037)
	Trust news or information from the US					
Russian language	0.017 (0.034)	-0.088 (0.082)	-0.013 (0.053)	0.032 (0.045)	0.032 (0.046)	0.019 (0.035)
	Trust news or information from Turkey					
Russian language	-0.090** (0.039)	-0.31*** (0.093)	-0.14** (0.065)	-0.062 (0.049)	-0.12** (0.051)	-0.089** (0.040)
	Trust news or information from China					
Russian language	0.068** (0.031)	-0.051 (0.073)	0.12** (0.049)	0.037 (0.040)	0.088** (0.043)	0.060* (0.032)
Observations	1,500	245	621	879	963	1,209
R-squared	0.076	0.174	0.138	0.076	0.100	0.062
Waves	After 2022	After 2022	After 2022	After 2022	After 2022	After 2022
Sample	All	Firm head	Above 40	Below 40	Informed	No Russians
Region FE	Yes	Yes	Yes	Yes	Yes	Yes

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Table shows the results of regression (14) for binary dependent variables, “Trust news or information from Russia/ the US/ Turkey/ China”. Robust standard errors in parentheses.

Figure D3: Event study design about favorable opinion of countries

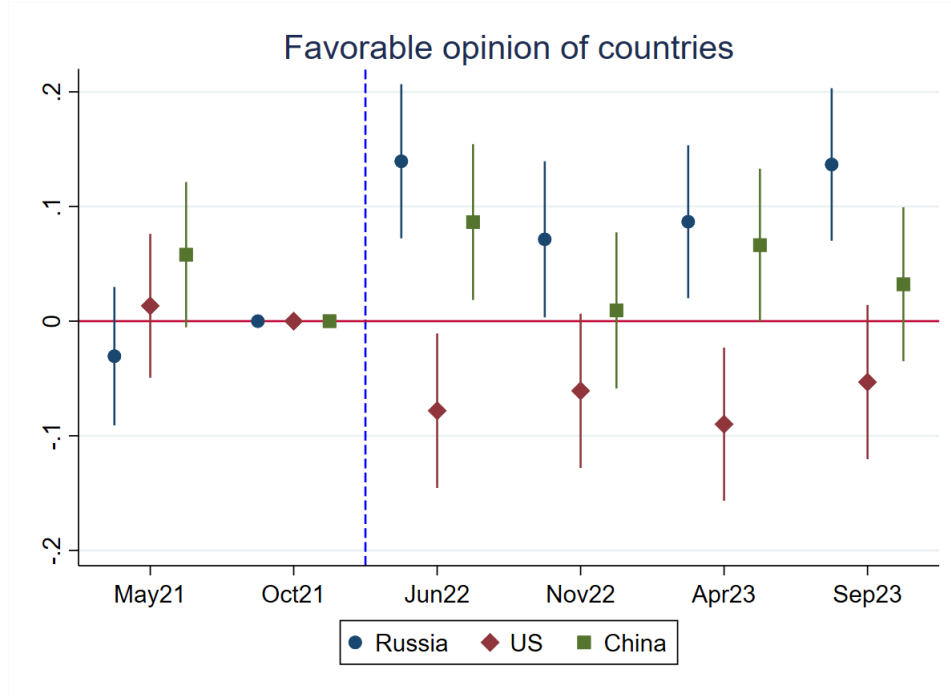
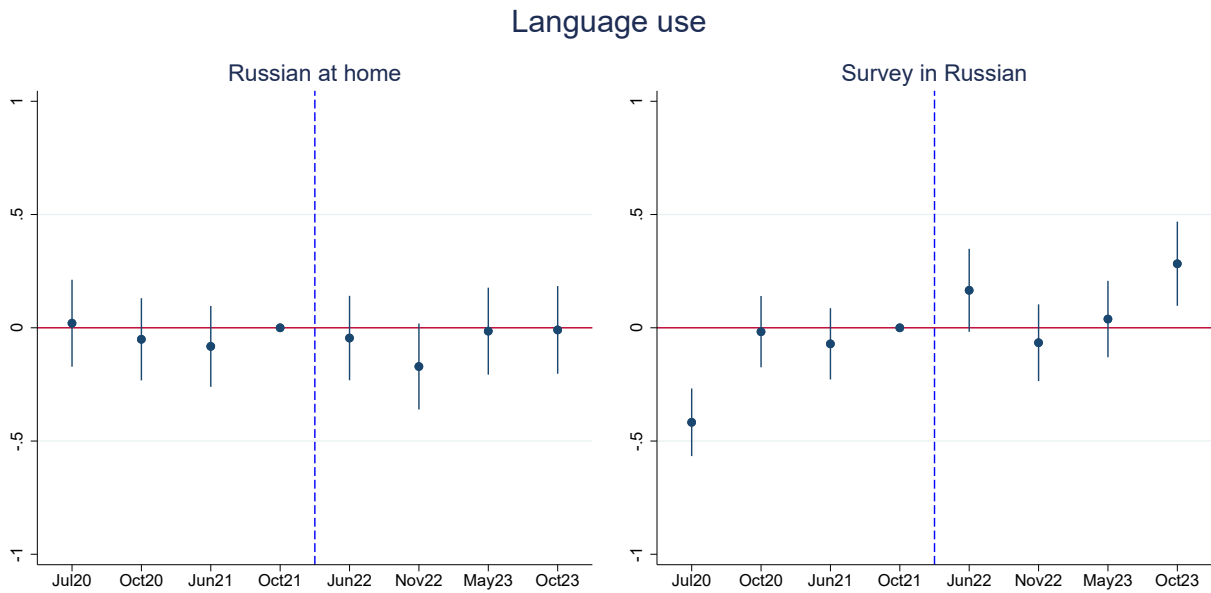


Figure D4: Event-study for Russian language use.



Note. The figure shows event-study coefficients γ_t for the DID specification (12) and dependent variables: “Russian at home” (left), “Survey in Russian” (right).

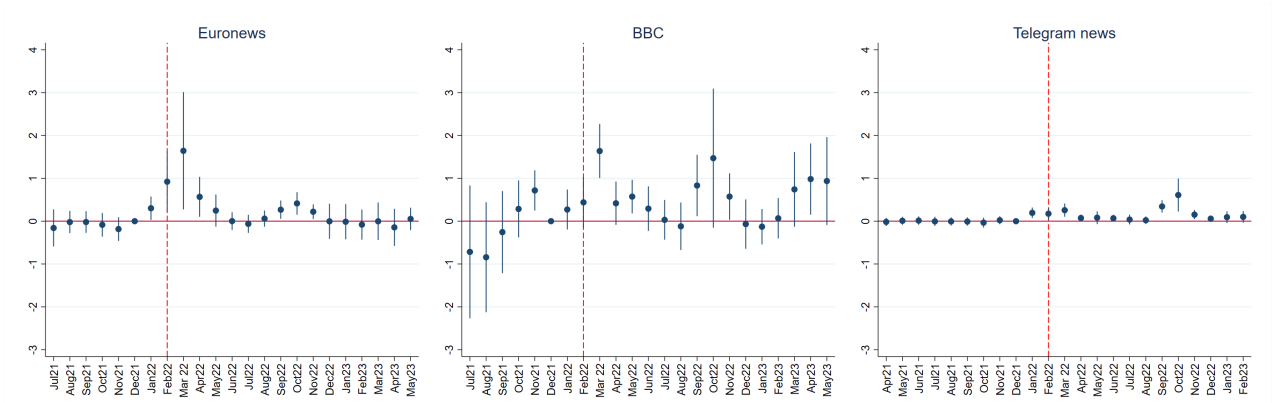
Table D14: Russian language use

VARIABLES	(1) Russian at home	(2)	(3) Survey in Russian	(4)
Treatment exposure (α)	0.62*** (0.033)		0.41*** (0.027)	
Post-invasion (β)	-0.0033 (0.019)		-0.18*** (0.020)	
Treatment exposure x Post-invasion (γ)	-0.023 (0.047)	-0.028 (0.047)	0.21*** (0.044)	0.22*** (0.043)
Constant	0.52*** (0.019)	0.75*** (0.038)	0.78*** (0.017)	0.97*** (0.040)
Observations	13,020	13,020	13,020	13,020
R-squared	0.490	0.512	0.228	0.291
Time FE	N	Y	N	Y
Region FE	N	Y	N	Y
Mean DV	.461	.461	.765	.765

Note. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The odd columns show results of regression (11), even columns – (12) for binary dependent variables: “Russian at home” (columns 1-2), “Survey in Russian” (columns 3-4). The treatment exposure is the regional share of Russian-speaking Kazakhs from Census 2021. The set of control variables includes gender, five age categories, a binary indicator for higher education, five categories of household size, a binary indicator for urban settlement, Russian and Kazakh binary indicators of ethnicity, as well as the interviewer’s assessment of the respondent’s survey comprehension and comfort with answering the survey.

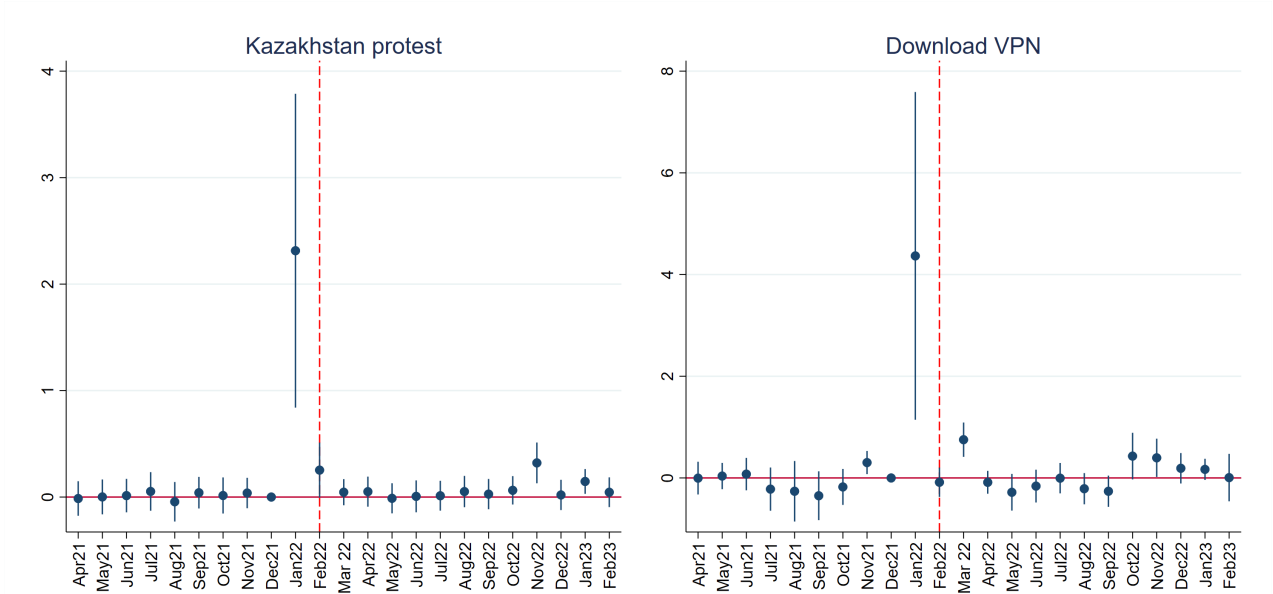
E Appendix

Figure E1: Event-study coefficients for Yandex search of Western outlets and Telegram news



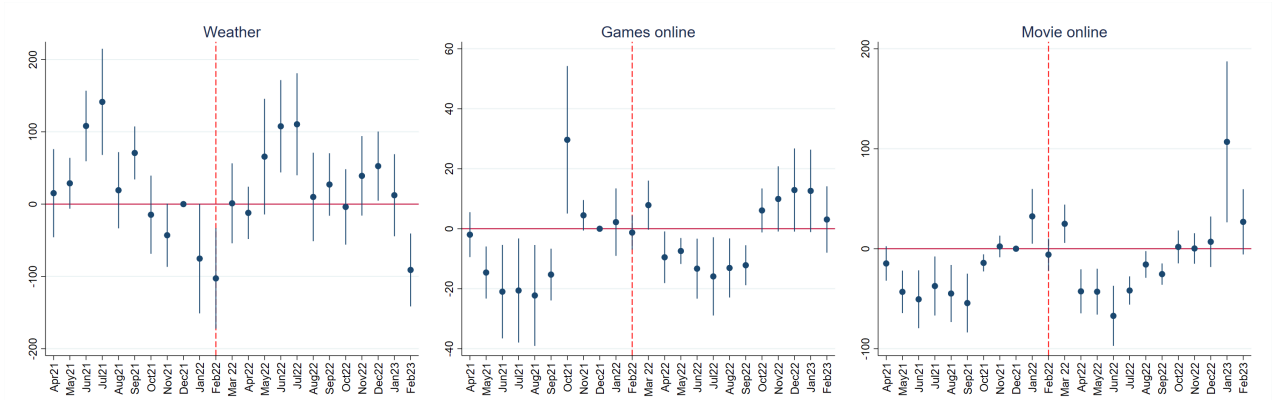
Note. The figure shows coefficients γ_τ from (13) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1000 regional inhabitants.

Figure E2: Event study design for Yandex search of January 2022 protest in Kazakhstan



Note. The figure shows coefficients γ_τ from (13) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs.

Figure E3: Event-study coefficients for Yandex search of weather and entertainment



Note. The figure shows coefficients γ_τ from (13) over time. The treatment exposure is the regional share of Russian-speaking Kazakhs. The outcome variable is the number of keyword searches of a keyword in Yandex per 1000 regional inhabitants.